

M13 Batch 6 — Full Set

M13.8 Instruments (ATA 31) · PDF pages 449–562

Altimeters, airspeed indicators, machmeters, vertical speed indicators, temperature systems, air data computers, ADIRU and standby instruments, stall warning.

326 questions · Level 3: 97 · Level 2: 166 · Level 1: 63

● marks a question built on a fact your instructor highlighted or underlined in the manual (267 of 326). Topics run in manual page order; inside each topic the questions are grouped by level, and the answers with explanations follow at the end of that topic.

No calculations, no unit conversions, no worked examples, nothing type-specific unless it was marked.

Altimeters and baro settings

PDF pages 449–461 · 43 questions

LEVEL 1 — RECALL

1. The quantity measured in the barometric method of determining altitude ●
 - a. Satellite signal delay
 - b. Radio wave delay
 - c. Air pressure
2. The quantity measured by a radio altimeter ●
 - a. Static air pressure
 - b. Radio wave delay
 - c. Gravitation model deviation
3. One of the factors on which the choice of altimeter depends ●
 - a. The transition level at the destination
 - b. The tolerance values required by law
 - c. The number of static ports fitted
4. The type of indication listed besides double pointer, triple pointer and drum-type ●
 - a. Altitude indication on the baro scale
 - b. Altitude indication on the transponder
 - c. Altitude indication on the PFD
5. The shape of the ISA pressure graph at low altitudes ●
 - a. Nearly linear
 - b. Strongly exponential
 - c. A series of steps
6. The pressure decrease that corresponds to an altitude increase of 30 ft at low altitude ●
 - a. 10 hPa
 - b. 30 hPa
 - c. 1 hPa
7. What the baro set knob gives the altimeter ●
 - a. A new reference pressure
 - b. A new static pressure source
 - c. A new scale calibration in metres

8. The crew is preparing for departure. Which pressure do they set on the altimeters? ●
 - a. The actual pressure related to sea level
 - b. The standard pressure of the ISA
 - c. The actual pressure at ground level
9. The name given to the reference pressure used for QNE ●
 - a. Ambient
 - b. Standard
 - c. Nautical
10. Who may select the barometric correction on an altimeter, besides the pilots ●
 - a. Maintenance personnel
 - b. Air traffic control
 - c. The air data computer

LEVEL 2 — UNDERSTAND AND EXPLAIN

11. How a barometer determines altitude ●
 - a. By using the decrease in pressure with increasing altitude
 - b. By using the increase in pressure with increasing altitude
 - c. By using the delay of a reflected radio wave
12. A method of determining altitude uses the WGS84 gravitation model as its reference instead of the ground. Which method is it?
 - a. Radio altimeter
 - b. GPS
 - c. Barometer
13. How the interior of a barometric altimeter casing is connected ●
 - a. To the outside air via the pitot system
 - b. To a sealed reference vacuum
 - c. To the outside air via the static system
14. An aircraft climbs. What happens inside its barometric altimeter? ●
 - a. Pressure decreases, the bellows contract and the pointer rises
 - b. Pressure decreases, the bellows expand and the pointer rises
 - c. Pressure increases, the bellows expand and the pointer falls
15. The reason for mounting 2 or 3 bellows in series in an altimeter ●
 - a. To allow the case to be sealed against static pressure
 - b. To balance the mechanism against acceleration
 - c. To obtain a more exact indication
16. An altitude indicator has no pneumatic connection at all, only wiring. Which technology is it?
 - a. An instrument with static pressure input
 - b. An indicator with electrical input from an air data computer
 - c. A drum-type instrument with static pressure input
17. The construction feature that distinguishes an encoding altimeter ●
 - a. Additional bellows and electrical connectors
 - b. An additional static port and a drum
 - c. Additional pointers and a second baro scale

18. An encoding altimeter puts out an ALT signal. Which system receives it? ●
- a. The ATC transponder system
 - b. The air data computer
 - c. The static system
19. Why is an encoding altimeter connected to the ATC transponder system? ●
- a. So that ground control can be informed about the altitude of the aircraft
 - b. So that the crew can be informed about the ground pressure
 - c. So that the transponder can set the baro correction of the altimeter
20. The aircraft in which encoding altimeters are found ●
- a. All aircraft fitted with an air data computer
 - b. Some small aircraft
 - c. Only aircraft without a transponder
21. The reason why an encoding altimeter is fitted with electrical connectors ●
- a. To provide the ALT signal for the ATC transponder system
 - b. To receive the reference pressure from an air data computer
 - c. To drive the additional bellows inside the instrument
22. How the baro set knob works in a pneumatic indicator ●
- a. It directly turns the whole altimeter mechanism
 - b. It moves the bellows stack alone
 - c. It sends a correction signal to the air data computer
23. A new reference pressure is selected on an electrical altitude indicator. How is the correction made? ●
- a. Electrically inside the indicator or via an air data computer
 - b. Mechanically, by turning the whole altimeter mechanism
 - c. Pneumatically, by an additional set of bellows
24. An airport lies well above sea level and the altimeter is set to QNH. What does the instrument show on the ground?
- a. The altitude of the airport related to mean sea level
 - b. The height of the aircraft above the runway
 - c. Zero feet
25. The aircraft is descending and reaches the Transition Level. What do the pilots do with the altimeters? ●
- a. They adjust them to 1013 hPa
 - b. They adjust them to the actual pressure at the destination airport
 - c. They adjust them to the pressure used at the departure airport
26. How a new reference pressure is applied in a pneumatic altimeter compared with an electrical indicator ●
- a. By turning the whole altimeter mechanism, compared with a correction made electrically inside the indicator or via an air data computer
 - b. By a correction made electrically inside the indicator or via an air data computer, compared with turning the whole altimeter mechanism
 - c. By turning the whole altimeter mechanism, in the pneumatic indicator and in the electrical indicator alike
27. The baro correction that makes the altimeter read zero on the ground ●
- a. QNE
 - b. QFE
 - c. QNH

28. The level to which the pressure used for QFE is related ●
- The standard atmosphere
 - Mean sea level
 - Ground level
29. An aircraft is in flight with QFE set on its altimeter. What does the indication represent? ●
- The height above mean sea level
 - The altitude in the standard atmosphere
 - The height above the ground
30. Why do pilots of medium and large aircraft normally not use QFE? ●
- The QFE value is not transmitted by data link
 - They can use a radio altimeter system
 - Their baro scales cannot be set that low
31. An aircraft in flight has the baro correction for QNH set. What does the altimeter read? ●
- The height above mean sea level
 - The height above the transition altitude
 - The height above the ground
32. A technician selects QNE on an altimeter. Which reference is then used? ●
- The ambient pressure at the airport
 - The ambient pressure at mean sea level on the day
 - The standard atmosphere with 1013 hPa
33. Where the QNE setting is normally used ●
- During takeoff and landing only
 - On the ground, not in flight
 - In flight, not on the ground
34. The advantage of QNE ●
- The altimeter reads zero at touchdown
 - Every aircraft reads its own height above the ground
 - All aircraft in cruise flight use the same reference pressure
35. What an altimeter in flight indicates with the baro correction for QFE compared with QNH ●
- The height above mean sea level with QFE and the height above the ground with QNH
 - The height above the ground with QFE and the height above mean sea level with QNH
 - The height above the ground with both corrections
36. The reason why an altimeter set to QFE indicates 0 ft at touchdown ●
- Its reference is the standard atmosphere with 1013 hPa
 - Its reference is the actual pressure related on mean sea level
 - Its reference is the actual pressure related on ground level

LEVEL 3 — APPLY AND REASON

37. An altimeter is set to standard pressure. Under which condition is its indicated altitude equal to the true altitude? ●
- When the pressure at sea level is really 1013 hPa
 - When the airport elevation is zero
 - When the aircraft is above the transition altitude

- 38.** A crew keeps 1013 hPa set on the altimeters for the whole cruise. Why does the indication not give the true altitude everywhere along the route? ●
- Because the indication is only correct where the pressure at sea level really is 1013 hPa
 - Because the pressure graph of the standard atmosphere is only linear at low altitudes
 - Because all aircraft in cruise flight select a different reference pressure
- 39.** Why do all aircraft in cruise use one common selected pressure instead of the local pressure? ●
- So that each of them indicates its true height above the ground
 - So that ground control can read the elevation of the airports
 - So that they all keep the same altitude separation in low and high pressure areas
- 40.** An altimeter is left calibrated to 1013 hPa while the real pressure at sea level below the aircraft is lower. What will the instrument indicate? ●
- The correct altitude
 - Too low an altitude
 - Too high an altitude
- 41.** An aircraft passes the Transition Altitude after takeoff and later the Transition Level before landing. Which setting is made at each point? ●
- The actual pressure at the departure airport at the Transition Altitude and 1013 hPa at the Transition Level
 - 1013 hPa at the Transition Altitude and the actual pressure at the destination airport at the Transition Level
 - 1013 hPa at both the Transition Altitude and the Transition Level
- 42.** Two altimeters at the same airport, which lies above sea level, read differently on the ground, one showing zero and the other the elevation of the airport. Which baro corrections are set? ●
- QFE on the one showing zero and QNE on the one showing the elevation
 - QNH on the one showing zero and QFE on the one showing the elevation
 - QFE on the one showing zero and QNH on the one showing the elevation
- 43.** The difference between the QNE and the QNH altimeter calibration, both of which refer the altitude to mean sea level ●
- QNE uses the actual pressure related to mean sea level while QNH uses the standard atmosphere with 1013 hPa
 - QNE uses the standard atmosphere with 1013 hPa while QNH uses the actual pressure related to mean sea level
 - QNE uses the actual pressure related to ground level while QNH uses the standard atmosphere with 1013 hPa

Answers — Altimeters and baro settings

- c** The barometer method measures air pressure; the other two quantities belong to the radio altimeter and to GPS. (L1, PDF p. 449)
- b** The radio altimeter works on the delay of the radio wave, not on pressure. (L1, PDF p. 449)
- b** Legal tolerance values are one of the three listed selection factors. (L1, PDF p. 453)
- c** The fourth kind of indication listed is the altitude display on the PFD. (L1, PDF p. 453)
- a** Close to the ground the pressure against altitude curve is nearly a straight line. (L1, PDF p. 457)
- c** The printed relationship at low altitude is 30 ft per 1 hPa. (L1, PDF p. 457)
- a** The knob only changes the reference pressure the indication is based on. (L1, PDF p. 459)
- a** Before takeoff the reference is the actual sea level pressure, which is the QNH setting. (L1, PDF p. 459)

9. b The 1013 hPa reference is called standard. (L1, PDF p. 461)
10. a The manual names pilots and maintenance personnel as the ones who select the baro correction. (L1, PDF p. 461)
11. a Pressure falls as altitude rises, and the barometer uses that relationship. (L2, PDF p. 449)
12. b WGS84 is the reference of the GPS method; the barometer uses pressure and the radio altimeter uses the ground. (L2, PDF p. 449)
13. c Outside air reaches the inside of the case through the static system. (L2, PDF p. 449)
14. b Falling pressure around the capsule lets the bellows expand, which drives the pointer up the scale. (L2, PDF p. 449)
15. c Several capsules in series give more movement and therefore a more exact indication. (L2, PDF p. 449)
16. b Without a static connection the altitude information can only come electrically from an air data computer. (L2, PDF p. 453)
17. a The encoding altimeter carries extra bellows and electrical connectors on top of the normal altimeter. (L2, PDF p. 455)
18. a The encoding altimeter feeds the ALT signal to the ATC transponder. (L2, PDF p. 455)
19. a The purpose of the ALT signal is to pass the aircraft altitude to ground control. (L2, PDF p. 455)
20. b The encoding altimeter is described as a fit found in some small aircraft. (L2, PDF p. 455)
21. a The electrical connectors carry the ALT signal that the ATC transponder system needs. (L2, PDF p. 455)
22. a In a pneumatic instrument the setting is made mechanically by turning the whole mechanism. (L2, PDF p. 459)
23. a Electrical indicators have no mechanism to turn, so the correction is done electrically. (L2, PDF p. 459)
24. a With sea level as the reference, an airport above sea level indicates its own elevation. (L2, PDF p. 459)
25. b At the Transition Level the standard setting is replaced by the destination setting. (L2, PDF p. 459)
26. a The pneumatic instrument is re-referenced mechanically, the electrical one electrically inside itself or through an air data computer. (L2, PDF p. 459)
27. b QFE uses the ambient pressure at ground level, so the indication on the ground is zero. (L2, PDF p. 461)
28. c QFE is the ambient pressure at ground level, unlike QNH which is related to mean sea level. (L2, PDF p. 461)
29. c With the ground as reference the instrument shows height above the ground. (L2, PDF p. 461)
30. b A radio altimeter already gives the height above the ground, so the QFE setting is not needed. (L2, PDF p. 461)
31. a QNH refers everything to mean sea level, in flight as well as on the ground. (L2, PDF p. 461)
32. c QNE is always the fixed standard atmosphere value, not a value measured on the day. (L2, PDF p. 461)
33. c QNE is the in-flight setting; on the ground the airport settings are used. (L2, PDF p. 461)
34. c The whole point of QNE is a single common reference for all aircraft in the cruise. (L2, PDF p. 461)
35. b QFE refers the indication to the ground and QNH to mean sea level. (L2, PDF p. 461)
36. c QFE takes the pressure at ground level as its reference, so the indication is zero on the ground. (L2, PDF p. 461)
37. a The conversion from static pressure to altitude only matches the true altitude when the real sea level pressure is the calibration value. (L3, PDF p. 457)
38. a Indicated altitude only equals true altitude where the real sea level pressure is 1013 hPa, and PDF page 464 adds that the altimeters cannot be reset for every point of the route. (L3, PDF p. 457)
39. c A common reference keeps the vertical separation between aircraft consistent whatever the local pressure is. (L3, PDF p. 459)
40. c With the reference set higher than the real sea level pressure, the instrument over-reads. (L3, PDF p. 459)
41. b The common standard pressure is taken at the Transition Altitude on the way up and given back for the local pressure at the Transition Level on the way down. (L3, PDF p. 459)
42. c QFE is referenced to the ground and reads zero there, while QNH is referenced to mean sea level and shows the airport elevation. (L3, PDF p. 461)
43. b Both refer the altitude to mean sea level, but QNE always takes the standard pressure while QNH takes the actual one. (L3, PDF p. 461)

Airspeed indicators

PDF pages 463–471 • 30 questions

LEVEL 1 — RECALL

44. The airspeed indicator is which type of measuring instrument ●
- a. A differential temperature meter
 - b. A differential pressure meter
 - c. An absolute pressure meter
45. The scale of the airspeed indicator is calibrated in ●
- a. Nautical miles
 - b. Knots
 - c. Kilometres per hour
46. VNE is ●
- a. The maximum allowable speed with flaps extended
 - b. The maximum allowed airspeed
 - c. The maximum airspeed for cruise flight
47. The green band on the airspeed indicator of a small aircraft shows the ●
- a. Normal Operating Range
 - b. Full Flap Operating Range
 - c. Caution range
48. On a helicopter airspeed indicator the red mark shows ●
- a. The range of transition lift
 - b. The maximum speed allowed for autorotation
 - c. The speed that must never be exceeded
49. On a helicopter airspeed indicator the red-white mark shows ●
- a. The maximum speed allowed for autorotation under all conditions
 - b. The speed that must never be exceeded
 - c. The range of transition lift
50. True airspeed is the speed with which the aircraft moves ●
- a. Through the surrounding air
 - b. Relative to the airspeed indicator scale
 - c. Over the ground
51. The speed which is shown on the indicator is called ●
- a. Ground speed
 - b. True airspeed
 - c. Indicated airspeed

LEVEL 2 — UNDERSTAND AND EXPLAIN

52. A technician traces the pitot line to the airspeed indicator capsule. Which pressure does that line deliver to the capsule interior? ●
- a. Static pressure from the static system
 - b. Total pressure from the pitot system
 - c. Dynamic pressure from the static system

53. The capsule of an airspeed indicator sits inside a sealed casing. Which pressure acts inside that casing? ●
- a. Static pressure from the static system
 - b. Dynamic pressure from the pitot system
 - c. Total pressure from the pitot system
54. In an airspeed indicator the difference between total pressure and static pressure gives ●
- a. Static pressure
 - b. Total pressure
 - c. Dynamic pressure
55. A technician is asked what the indicated airspeed value is used for. Which answer is correct?
- a. As a value for thrust and for the maximum cabin pressure
 - b. As a value for altitude and for the maximum rate of climb
 - c. As a value for lift and for the maximum load on the flaps
56. An aircraft is fitted with a primary flight display instead of a mechanical airspeed indicator. How is airspeed presented? ●
- a. As a speed band on the PFD
 - b. As a speed bug on the PFD
 - c. As a speed limit marking on the PFD
57. In small aircraft the airspeed limits are shown by
- a. A movable pointer driven by the air data computer
 - b. A digital readout beside the airspeed indicator
 - c. Coloured ranges and lines on the airspeed indicator scale
58. An aeroplane is being flown with the airspeed in the yellow band. Which condition applies? ●
- a. It may be used in any air, including hard manoeuvres
 - b. It may only be used in smooth air without hard manoeuvres
 - c. It may only be used with the flaps extended
59. The slight increase of VMO with altitude is the result of taking account of ●
- a. The compressibility error of the IAS
 - b. The instrument error of the IAS
 - c. The static source error of the IAS
60. Airspeed indicators of jet aircraft do not show the low speed limit because ●
- a. It is already shown by the maximum speed pointer
 - b. It is a fixed value and therefore needs no indication
 - c. It depends on many factors such as weight and flap position
61. A crew wants a minimum speed indication in the cockpit. On which aircraft is it available, and where is it shown? ●
- a. Only on aircraft with a glass cockpit, on the speed tape of the PFD
 - b. On all jet aircraft, as a coloured band on the airspeed indicator scale
 - c. Only on piston engine aircraft, as a red line on the airspeed indicator

62. How the yellow band on the airspeed indicator of a small aircraft differs from the green band ●
- It is the normal operating range, while the green band is the caution range, to be used only in smooth air without hard manoeuvres
 - It is the caution range, to be used only in smooth air without hard manoeuvres, while the green band is the normal operating range
 - It is the caution range, to be used only in smooth air without hard manoeuvres, while the green band is the full flap operating range
63. A helicopter airspeed indicator carries two essential marked limits. What makes them necessary?
- The aerodynamic properties of the helicopter
 - The engine power limits of the helicopter
 - The maximum take-off weight of the helicopter
64. An aircraft is flown at sea level in standard conditions and its IAS equals its TAS. Which further condition must be met? ●
- With negligible instrument error
 - With negligible static source error
 - With negligible compressibility error
65. TAS is obtained from IAS by correcting for instrument error, static source error, dynamic pressure error and ●
- Aircraft weight
 - Magnetic variation
 - Air density
66. The air density correction applied to airspeed depends on ●
- Pressure and temperature
 - Temperature and aircraft weight
 - Pressure and humidity

LEVEL 3 — APPLY AND REASON

67. A jet aeroplane climbs from low altitude to high altitude. Which limit governs its maximum speed at the higher altitude? ●
- The never exceed speed VNE
 - The maximum allowed Mach number MMO
 - The maximum operating velocity VMO
68. Why can the speed limit of a jet aeroplane at low altitude be read directly as an indicated airspeed? ●
- Because the limit there depends on dynamic pressure, and the indicated airspeed indicator is a dynamic pressure indicator
 - Because the limit there depends on dynamic pressure, and the indicated airspeed indicator is a total pressure indicator
 - Because the limit there depends on the speed of sound, and the indicated airspeed indicator is a dynamic pressure indicator
69. The change of VMO and of MMO as a jet aeroplane climbs ●
- VMO stays constant or increases slightly while MMO decreases
 - VMO and MMO both decrease
 - VMO decreases while MMO stays constant or increases slightly

70. A helicopter airspeed indicator carries a red mark and a red-white mark. Which limit does each of them give? ●
- The red mark the speed that must never be exceeded, the red-white mark the maximum speed allowed for autorotation
 - The red mark the speed that must never be exceeded, the red-white mark the range of transition lift
 - The red mark the maximum speed allowed for autorotation, the red-white mark the speed that must never be exceeded
71. An aircraft climbs while holding a constant indicated airspeed. What happens to the difference between IAS and TAS? ●
- It decreases, because the air density decreases
 - It stays the same, because the IAS is corrected for density
 - It increases, because the air density decreases
72. An aircraft is flown at sea level in standard conditions, but its static source error is not negligible. Do the indicated airspeed and the true airspeed coincide? ●
- Yes, because the static source error only separates them at altitude
 - Yes, because at sea level the two speeds always coincide
 - No, because they coincide only when the static source error is negligible as well
73. An aircraft is not fitted with an air data computer. What must be provided instead?
- Correction tables in the aircraft technical log
 - Correction tables in the Flight Operations Manual
 - A second airspeed indicator for comparison

Answers — Airspeed indicators

44. **b** The instrument works on the difference between two pressures, so it is a differential pressure meter. (L1, PDF p. 463)
45. **b** The printed text gives knots as the unit of the scale. (L1, PDF p. 463)
46. **b** VNE is the never exceed speed, which is the maximum allowed airspeed. (L1, PDF p. 465)
47. **a** Green marks the normal operating range on the scale. (L1, PDF p. 465)
48. **c** Red marks VNE, the never exceed speed. (L1, PDF p. 467)
49. **a** The red-white mark is the autorotation speed limit. (L1, PDF p. 467)
50. **a** TAS is defined as the aircraft speed through the surrounding air mass. (L1, PDF p. 469)
51. **c** What the instrument displays is the indicated airspeed. (L1, PDF p. 469)
52. **b** The capsule interior receives total pressure, which the pitot system delivers. (L2, PDF p. 463)
53. **a** The casing is fed from the static system, so static pressure surrounds the capsule. (L2, PDF p. 463)
54. **c** Total pressure minus static pressure is by definition the dynamic pressure. (L2, PDF p. 463)
55. **c** The manual lists lift and maximum load on the flaps and airframe among the uses of the IAS. (L2, PDF p. 463)
56. **a** The third type of airspeed indication listed is the speed band shown on the PFD. (L2, PDF p. 463)
57. **c** Small aircraft carry the limits as colour markings painted on the instrument scale. (L2, PDF p. 465)
58. **b** The caution range is restricted to smooth air with no hard manoeuvres. (L2, PDF p. 465)
59. **a** The manual attributes the slight rise to the compressibility error of the IAS. (L2, PDF p. 465)
60. **c** The low speed limit varies with weight and flap position, so it is not marked on the dial. (L2, PDF p. 465)
61. **a** The minimum speed indication is a glass cockpit feature shown on the PFD speed tape. (L2, PDF p. 465)
62. **b** The green band is the normal operating range while the yellow band is a caution range with conditions attached to its use. (L2, PDF p. 465)
63. **a** The manual gives the aerodynamic properties of the helicopter as the reason for the two marked limits. (L2, PDF p. 467)

- 64. **b** The manual gives two conditions for IAS and TAS to coincide: standard conditions at sea level and negligible static source error. (L2, PDF p. 469)
- 65. **c** Air density is the fourth correction listed for deriving TAS from IAS. (L2, PDF p. 471)
- 66. **a** The manual states air density depends on pressure and temperature. (L2, PDF p. 471)
- 67. **b** Above a certain altitude the Mach limit MMO takes over from the dynamic pressure limit VMO. (L3, PDF p. 465)
- 68. **a** This page prints that the low altitude limit is a dynamic pressure limit, and PDF page 468 prints that the indicated airspeed is a dynamic pressure indicator, so the limit can be marked as a speed. (L3, PDF p. 465)
- 69. **a** VMO holds or rises a little with altitude while MMO falls, because the speed of sound falls. (L3, PDF p. 465)
- 70. **a** The plain red mark is the never exceed speed and the red-white mark is the autorotation limit. (L3, PDF p. 467)
- 71. **c** Falling air density with altitude widens the gap between IAS and TAS. (L3, PDF p. 469)
- 72. **c** The printed condition is both standard conditions at sea level and a negligible static source error, so failing one of them is enough to separate the two speeds. (L3, PDF p. 469)
- 73. **b** Without an air data computer the corrections come from tables in the Flight Operations Manual. (L3, PDF p. 471)

Machmeters and overspeed warning

PDF pages 473–479 · 23 questions

LEVEL 1 — RECALL

74. What is the purpose of the mach meter ●
- To enable flight at the maximum operating mach number and with economic efficiency
 - To enable flight at the maximum operating velocity and with minimum structural load
 - To enable flight at the never exceed speed and with maximum rate of climb
75. A technician opens a mach meter on the bench. Which two sensing elements are inside it? ●
- Aneroid bellows and a capillary restrictor
 - Differential pressure bellows and a bimetallic element
 - Differential pressure bellows and aneroid bellows
76. The mach number M is defined as the relation of ●
- Speed flown to speed of sound
 - True airspeed to calibrated airspeed
 - Speed of sound to speed flown
77. Which errors is the indicated mach number M_i subject to ●
- Instrument error and compressibility error
 - Instrument error and static source error
 - Static source error and density error
78. In a jet aircraft the speed limit is shown in the airspeed indicator by ●
- A red and white VNE pointer
 - A red and white VMO / MMO pointer
 - A red and green VMO / MMO pointer
79. On the speed tape of a PFD the speed limit is shown by ●
- An amber and black area
 - A red and white area
 - A red and black area

LEVEL 2 — UNDERSTAND AND EXPLAIN

80. The reason why a mach indication is needed for flight at high altitude ●
- Because the limit there is the maximum operating mach number and the speed of sound is a function of altitude
 - Because the limit there is the maximum operating mach number and the speed of sound is a function of dynamic pressure
 - Because the limit there is the maximum operating velocity and dynamic pressure is a function of altitude
81. What does the shock formation caused by compression of the air do to the airflow and the forces on the aircraft ●
- It prevents laminar airflow, increases drag and decreases lift
 - It prevents laminar airflow, decreases drag and decreases lift
 - It prevents turbulent airflow, decreases drag and increases lift

- 82.** Why can the speed of sound be represented as a function of altitude ●
- Because it is a function of static pressure, and under standard conditions pressure is a function of temperature
 - Because it is a function of density, and under standard conditions density is a function of airspeed
 - Because it is a function of temperature, and under standard conditions temperature is a function of altitude
- 83.** The MMO marking on a stand-alone mach meter is the adjustable version and has to be re-set. Where is this carried out?
- On the ramp before each flight
 - In the workshop
 - In the cockpit by the pilot
- 84.** How the indicated mach number M_i is obtained compared with the true mach number M_T ●
- M_i is calculated from pitot pressure and dynamic pressure while M_T is calculated by the air data computer
 - M_i is calculated by the air data computer while M_T is calculated from pitot pressure and dynamic pressure
 - M_i is calculated from pitot pressure and dynamic pressure while M_T is calculated from the speed of sound and the outside temperature
- 85.** From which pressures is the indicated mach number M_i calculated ●
- Dynamic pressure and cabin pressure
 - Pitot pressure and accumulator pressure
 - Pitot pressure and dynamic pressure
- 86.** The true mach number M_T is calculated by ●
- The air data computer
 - The flight management computer
 - The central maintenance computer
- 87.** Two mach values are available on an aircraft, one free of instrument and static source error. Which value is that? ●
- The true mach number M_T
 - The indicated mach number M_i
 - The maximum operating mach number MMO
- 88.** Why do jet aircraft need a pointer or speed tape marking to show the maximum allowed airspeed
- Because the speed limit is a fixed value as the VNE is in piston engine aircraft
 - Because the airspeed indicator cannot show indicated airspeed above the speed of sound
 - Because the speed limit is not a fixed value as the VNE is in piston engine aircraft
- 89.** An aircraft approaches its speed limit at low altitude. What sets that limit? ●
- The maximum allowed mach number
 - The static pressure which acts on the aircraft structure
 - The dynamic pressure which acts on the aircraft structure
- 90.** How does VMO behave with increasing altitude when the compressibility error of the IAS is taken into account ●
- It always decreases in the same way as MMO
 - It is either constant or it decreases slightly
 - It is either constant or it increases slightly

91. An aircraft climbs steadily and the MMO limit falls. Why does it fall? ●
- Because the compressibility error increases
 - Because the speed of sound decreases
 - Because the dynamic pressure decreases
92. Why do all jet aircraft have an independent aural overspeed warning
- Because the static source error becomes too large at high altitude
 - Because the airspeed indicator is unreadable at high mach numbers
 - Because of the danger of aircraft damage during an overspeed
93. How is the aural overspeed warning produced ●
- From the cockpit loudspeakers triggered by the master warning system, or by a separate system with a horn from the stall warning
 - From the cockpit loudspeakers triggered by the master warning system, or by a separate system with a clacker sound
 - From the cockpit loudspeakers triggered by the air data computer, or by a separate system with a clacker sound

LEVEL 3 — APPLY AND REASON

94. A machmeter is opened and found to contain a differential pressure bellows and an aneroid bellows. Which instruments use these same two sensing elements? ●
- The airspeed indicator uses a differential pressure capsule and the vertical speed indicator an aneroid capsule
 - The altimeter uses a differential pressure capsule and the airspeed indicator an aneroid capsule
 - The airspeed indicator uses a differential pressure capsule and the altimeter an aneroid capsule
95. A jet aeroplane in high altitude cruise triggers its overspeed warning. Which limit has been exceeded, and does the warning depend on the airspeed indication? ●
- MMO, and the detection is taken from the airspeed indication
 - MMO, and the detection is independent of the airspeed indication
 - VMO, and the detection is independent of the airspeed indication
96. The airspeed indication in the cockpit has failed. What does this mean for overspeed detection? ●
- Detection continues, because it is always independent of the airspeed indication
 - Detection is lost, because it is taken from the airspeed indication
 - Detection continues only while the aircraft is below the tropopause

Answers — Machmeters and overspeed warning

74. a The manual gives exactly these two purposes for the mach meter. (L1, PDF p. 473)
75. c The manual lists these two elements as the contents of a mach meter. (L1, PDF p. 473)
76. a Mach number is speed flown divided by speed of sound. (L1, PDF p. 475)
77. b These are the two errors stated for Mi. (L1, PDF p. 475)
78. b The manual names the red and white VMO / MMO pointer. (L1, PDF p. 479)
79. c The PFD speed tape uses a red and black area. (L1, PDF p. 479)
80. a PDF page 470 prints that at higher altitudes the limit is the maximum operating mach number, and this page prints that the speed of sound is a function of altitude, so the limit cannot be read from an airspeed alone. (L2, PDF p. 473)
81. a The page states shock formation stops laminar flow, raises drag and lowers lift. (L2, PDF p. 473)
82. c The chain given is speed of sound to temperature to altitude. (L2, PDF p. 473)
83. b The manual calls adjusting the MMO marking a workshop task. (L2, PDF p. 473)

- 84. a Mi comes from the pressures at the instrument while MT is worked out by the air data computer. (L2, PDF p. 475)
- 85. c The manual names pitot pressure and dynamic pressure as the inputs for Mi. (L2, PDF p. 475)
- 86. a The manual states the air data computer calculates MT. (L2, PDF p. 475)
- 87. a MT is defined as being without instrument and static source error, unlike Mi. (L2, PDF p. 475)
- 88. c The limit varies, so it must be displayed, unlike a fixed VNE. (L2, PDF p. 477)
- 89. c VMO is a structural limit set by dynamic pressure. (L2, PDF p. 477)
- 90. c The manual states VMO is constant or rises slightly with altitude. (L2, PDF p. 477)
- 91. b The falling speed of sound is the stated reason. (L2, PDF p. 477)
- 92. c The stated reason is the risk of damaging the aircraft. (L2, PDF p. 479)
- 93. b These are the two arrangements given for the aural warning. (L2, PDF p. 479)
- 94. c PDF page 468 prints that the airspeed indicator has a differential pressure capsule and PDF page 454 that the altimeter's measuring component is an aneroid capsule, which are the two bellows listed here. (L3, PDF p. 473)
- 95. b PDF page 470 prints that at higher altitudes the limit is the maximum allowed mach number, and this page prints that the overspeed detection is always independent of the airspeed indication. (L3, PDF p. 479)
- 96. a Detection is stated to be always independent of the airspeed indication. (L3, PDF p. 479)

Vertical speed indicators

PDF pages 481–487 · 20 questions

LEVEL 1 — RECALL

97. What does the variometer measure in order to display vertical speed ●
- Air temperature changes
 - Air pressure changes
 - Vertical acceleration forces
98. In which units is the variometer scale calibrated ●
- Knots or m/sec
 - Ft/sec or m/min
 - Ft/min or m/sec
99. What does the abbreviation IVSI stand for ●
- Inertial Vertical Speed Indicator
 - Instantaneous Vertical Speed Indicator
 - Integrated Vertical Speed Indicator
100. What does the IVSI indicator use to produce its undelayed indication ●
- A small pump with a magnetically held piston
 - A small fan with a spring loaded vane
 - A small pump with a spring loaded piston
101. A diaphragm variometer is stripped for overhaul. What seals the air gap? ●
- A rubber lip seal
 - Badger hair
 - A felt washer
102. What sets the central position of a diaphragm variometer ●
- A permanent magnet
 - A coiled spring
 - A hairspring driven by the capillary

LEVEL 2 — UNDERSTAND AND EXPLAIN

103. The reason why a vertical speed indicator is connected to the static system only ●
- Because it works from changes of static pressure acting on its bellows
 - Because it works from the delay of the total pressure through a capillary
 - Because it works from the difference between total pressure and static pressure
104. The reason why the outside of the variometer bellows is fed through a capillary ●
- So that the same pressure always acts inside and outside the bellows
 - So that the pressure change outside the bellows is delayed
 - So that the pressure change inside the bellows is delayed
105. How are the two sides of the differential pressure bellows of a mechanical variometer supplied ●
- The outside directly with static pressure, the inside delayed via a capillary
 - The inside directly with static pressure, the outside delayed via a capillary
 - The inside directly with total pressure, the outside delayed via a capillary

- 106.** An aircraft holds a constant altitude so static pressure stays constant. What does the mechanical variometer show?
- Zero, because the same pressure acts inside and outside the bellows
 - A small descent, because the bellows remain partly compressed
 - A small climb, because the capillary keeps the outside pressure higher
- 107.** An aircraft with an IVSI enters and leaves a turn with a high bank angle. What should be expected? ●
- An inaccurate vertical speed indication, because any vertical acceleration affects the piston
 - A total loss of indication, because the piston is locked by the acceleration
 - A correct vertical speed indication, because the piston compensates for bank angle
- 108.** What produces the force that moves the diaphragm and the pointer of a diaphragm variometer
- The difference between static pressure and total pressure
 - The difference between accumulator pressure and cabin pressure
 - The difference between static pressure and accumulator pressure
- 109.** How is pressure balanced out in a diaphragm variometer ●
- Via a capillary tube between the diaphragm and the casing
 - Via the air gap between the diaphragm and the casing
 - Via a calibrated orifice in the centre of the diaphragm
- 110.** A precision variometer is required for a glider. Why is the diaphragm type chosen? ●
- Because diaphragm variometers are exact
 - Because diaphragm variometers need no static source
 - Because diaphragm variometers have the longest time constant

LEVEL 3 — APPLY AND REASON

- 111.** Why is the pointer movement of a mechanical variometer delayed when the vertical speed changes
- Because the change is immediate inside the bellows but delayed outside by capillary action and the larger volume
 - Because the pointer linkage is deliberately damped by a friction brake
 - Because the change is immediate outside the bellows but delayed inside by capillary action and the larger volume
- 112.** The response of a mechanical variometer to a change of vertical speed compared with an instantaneous vertical speed indicator
- Delayed in the mechanical variometer, and undelayed in the IVSI because an acceleration piston is installed
 - Undelayed in the mechanical variometer, and delayed in the IVSI because an acceleration piston is installed
 - Delayed in both, because both feed the outside of the bellows through a capillary
- 113.** An aircraft begins a descent. How does the IVSI piston give an immediate indication?
- The negative gravity force closes the capillary so that the outside static pressure changes first
 - The negative gravity force moves the piston down and lowers the pressure inside the bellows before the outside static pressure changes
 - The negative gravity force moves the piston up and raises the pressure inside the bellows before the outside static pressure changes

- 114.** A thermometer must read down to minus 150 C. Which filling substance has an application range that reaches this?
- Mercury
 - Toluene
 - Pentane
- 115.** A mercury thermometer is offered for work at minus 60 C. Why is it the wrong choice?
- Its application range only starts at minus 30 C
 - It allows no remote temperature measurement
 - Its expansion coefficient is temperature-dependent
- 116.** A filling is wanted that reads up to 1000 C without a poison hazard. Which substance fits, and what must be accepted with it?
- Mercury, with heavy wetting of the thermometer tube
 - Galinstan, with a health risk from inflammable vapour
 - Galinstan, with heavy wetting of the thermometer tube

Answers — Vertical speed indicators

- 97. b** The variometer works from changes in air pressure. (L1, PDF p. 481)
- 98. c** The manual gives ft/min or m/sec. (L1, PDF p. 481)
- 99. b** IVSI is the instantaneous vertical speed indicator, giving undelayed indication. (L1, PDF p. 483)
- 100. c** The manual describes a small pump with a spring loaded piston. (L1, PDF p. 483)
- 101. b** The manual states the gap is sealed with badger hair. (L1, PDF p. 485)
- 102. b** A coiled spring sets the central position. (L1, PDF p. 485)
- 103. a** The instrument senses air pressure changes with static pressure on both sides of its bellows, so it needs no pitot connection. (L2, PDF p. 481)
- 104. b** The capillary holds back the pressure change on the outside, which is what creates the difference the pointer shows. (L2, PDF p. 481)
- 105. b** Static pressure reaches the inside directly and the outside through the capillary. (L2, PDF p. 481)
- 106. a** With equal pressures either side of the bellows the pointer reads zero. (L2, PDF p. 481)
- 107. a** Any vertical acceleration acts on the piston, so turn entry and exit give a wrong reading. (L2, PDF p. 483)
- 108. c** The driving difference is between static and accumulator pressure. (L2, PDF p. 485)
- 109. b** Balancing takes place through the air gap between diaphragm and casing. (L2, PDF p. 485)
- 110. a** Their accuracy is the stated reason for their use as precision instruments. (L2, PDF p. 485)
- 111. a** The capillary and the larger outer volume delay the outside pressure change. (L3, PDF p. 481)
- 112. a** PDF page 486 prints that the pointer movement is delayed on a change of vertical speed, and this page prints that the IVSI removes that delay with an acceleration piston. (L3, PDF p. 483)
- 113. c** The piston raises bellows pressure ahead of the real static pressure change. (L3, PDF p. 483)
- 114. c** Only pentane, listed from minus 200 C, covers that low a reading; toluene stops at minus 80 C and mercury at minus 30 C. (L3, PDF p. 487)
- 115. a** Mercury is only listed from minus 30 C upwards, so minus 60 C is outside its range. (L3, PDF p. 487)
- 116. c** Galinstan is the non-poisonous alloy listed up to 1000 C, and its drawback is heavy wetting of the tube. (L3, PDF p. 487)

Temperature-indicating systems

PDF pages 489–495 • 36 questions

LEVEL 1 — RECALL

- 117.** The airspeed value that the temperature display is used to determine ●
- TAS
 - IAS
 - EAS
- 118.** The hazard that the temperature display helps the crew detect ●
- Possible pitot blockage
 - Possible engine surge
 - Possible ice formation
- 119.** What static air temperature (SAT) is ●
- The ambient temperature plus the rise from air compression
 - The actual temperature of the ambient air
 - The free airstream temperature with a built-in deflection error
- 120.** What total air temperature (TAT) is ●
- The ambient temperature minus the temperature generated by air compression
 - The ambient temperature plus the temperature generated by air compression
 - The ambient temperature with no allowance for air compression
- 121.** The temperature value that is determined directly by the measuring probe ●
- TAT
 - RR
 - SAT
- 122.** What a TAT measuring probe contains as its sensing elements ●
- Two or three temperature dependent capillary tubes
 - One temperature dependent PTC resistor
 - Two or three temperature dependent PTC resistors

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 123.** The engine related use of the temperature display ●
- Calculating engine performance and determining engine limits
 - Calculating engine oil quantity and determining oil limits
 - Calculating engine vibration and determining balance limits
- 124.** What an electronic OAT display needs in order to show a reading ●
- An external OAT sensor
 - An internal cabin thermistor
 - A direct reading bulb in the cockpit
- 125.** The aircraft on which the simple OAT indicating system is used ●
- Small propeller driven aeroplanes and helicopters
 - Large commercial jet aeroplanes
 - Supersonic transport aeroplanes

- 126.** A fast jet aeroplane cannot use the simple OAT indicating system. What is the reason? ●
- It needs a bleed air supply that jets cannot provide
 - It is only accurate up to an airspeed of 250 kts
 - It cannot be protected against ice
- 127.** In a modern temperature indicating system, the display and the calculation ●
- are performed together in the instrument, as in conventional analogue construction
 - are performed separately, the calculation being done in the computer
 - are performed separately, the calculation being done in the measuring probe
- 128.** How SAT is obtained in a fast aeroplane ●
- It is read directly from the total air temperature probe
 - It is read directly from a flush skin probe
 - It is calculated by an air data computer
- 129.** Why the TAT is always higher than the SAT during flight ●
- The probe heater adds heat to the measurement in flight
 - Ram air is present at the measuring probe and raises its temperature
 - The probe is flush with a skin that is warmed by the sun
- 130.** The built-in error that makes RAT differ from TAT ●
- The air is deflected as it flows around the aircraft fuselage
 - The air is warmed by the probe anti-icing element
 - The air is brought completely to rest inside the probe neck
- 131.** The form of temperature supplied by a probe mounted flush with the aircraft skin ●
- SAT
 - TAT
 - RAT
- 132.** A temperature probe is carried on a spacing foot clear of the fuselage skin. What is this arrangement for? ●
- To keep the probe clear of the skin heating system
 - To let bleed air be blown back over the skin
 - To determine the total air temperature out in the airstream
- 133.** Where the switching sits in an analogue temperature indicator of conventional construction ●
- In the air data computer, together with the measuring resistor
 - Directly in the instrument, except for the measuring resistor
 - Directly in the probe, except for the measuring resistor
- 134.** Besides the display instrument and the flight recorder, which user is supplied with the outside air temperature?
- The stall warning computer, to set the shaker threshold
 - The ADC or FMC, to calculate values that cannot be measured directly
 - The cabin pressure controller, to set the cabin rate of climb
- 135.** What produces the temperature rise seen at the measuring probe in flight ●
- Compression of the air
 - Radiation from the warm aircraft skin
 - Friction in the probe heating element

- 136.** The value on which the ram rise depends ●
- The pressure altitude
 - The mach number
 - The static pressure
- 137.** Why a ram effect occurs inside the TAT probe ●
- The air is warmed by the resistor heads before it leaves at the neck
 - The air is slowed right down as it passes the resistor heads and leaves at the neck
 - The air is speeded up as it passes the resistor heads and leaves at the neck
- 138.** The behaviour of the TAT probe heating in flight and on the ground
- In flight it raises the measurement, and on the ground it is often switched off automatically
 - In flight it has no effect on the measurement, and on the ground it is switched on automatically
 - In flight it has no effect on the measurement, and on the ground it is often switched off automatically
- 139.** What an aspirated TAT probe is supplied with ●
- Conditioned cabin air
 - Pitot pressure air
 - Bleed air
- 140.** The measuring wire of a TAT probe is broken. Which indication results, and why? ●
- The minus 70 C lower stop, because the resistance is infinitely high
 - The plus 50 C upper stop, because the resistance is infinitely high
 - The plus 50 C upper stop, because the resistance falls to zero
- 141.** The measuring wire of a TAT probe is short circuited. Which indication results, and why? ●
- The minus 70 C lower stop, because the resistance is 0 ohm
 - The plus 50 C upper stop, because the resistance is 0 ohm
 - The minus 70 C lower stop, because the resistance is infinitely high
- 142.** How the resistance of the TAT probe sensing elements changes as the sensed temperature rises ●
- It decreases
 - It stays constant
 - It increases

LEVEL 3 — APPLY AND REASON

- 143.** In flight, how do the three measured air temperature values compare with one another? ●
- SAT is the highest, RAT is lower than SAT but higher than TAT
 - RAT is the highest, TAT is lower than RAT but higher than SAT
 - TAT is the highest, RAT is lower than TAT but higher than SAT
- 144.** A flush skin probe supplies temperature in the RAT form. Why can that reading not be used as the ambient air temperature? ●
- RAT is deflected around the fuselage, so it reads lower than the ambient SAT
 - RAT still contains part of the ram rise, so it reads higher than the ambient SAT
 - RAT is taken downstream of the probe heater, so it reads higher than the ambient SAT
- 145.** Why does a fast aeroplane need an air data computer for ambient air temperature while a light propeller aeroplane manages with a simple thermometer?
- At high speed the probe heating is permanently on, so the measured value must be corrected
 - At high speed the ambient air temperature falls below the range of a direct reading thermometer
 - At high speed the ram rise is large, so the ambient SAT must be calculated from the measured value

- 146.** An aeroplane accelerates to a higher mach number while the static air temperature stays the same. What happens to the gap between TAT and SAT? ●
- It grows, because the ram rise depends on the mach number
 - It stays the same, because the ram rise depends only on the static air temperature
 - It shrinks, because the probe recovers less of the airflow at high speed
- 147.** In cruise the crew needs the static air temperature. How is that value produced? ●
- It is read directly from a second, unheated static temperature probe
 - It is read directly from the ram air temperature indicator
 - It is calculated from the TAT and the mach number in the air data computer
- 148.** The measuring wire of a TAT probe goes open circuit. Why does the indication run hard against a stop instead of simply reading low? ●
- Temperature is measured by a resistor bridge, and an infinitely high resistance drives the indication to the -70 C lower stop
 - Temperature is measured by a thermocouple, and the loss of the junction drives the indication to the +50 C upper stop
 - Temperature is measured by a resistor bridge, and an infinitely high resistance drives the indication to the +50 C upper stop
- 149.** The air in a TAT probe is slowed at the resistor heads to the point where a ram effect occurs. The temperature value the probe then supplies, and the way the ambient value is obtained from it
- TAT, and the SAT is calculated from it in the air data computer
 - SAT, and the TAT is calculated from it in the air data computer
 - RAT, and the SAT is calculated from it in the air data computer
- 150.** An aeroplane stands in strong sunshine and its skin is hot. Which probe design keeps these conditions out of the temperature reading, and how? ●
- A standard TAT probe, because its heating element masks the effect of the sun
 - An aspirated TAT probe, because compressed air blown backwards carries outside air along
 - A flush skin probe, because the skin and the probe then settle at the same temperature
- 151.** The heating of a TAT probe has failed and the aircraft is in icing conditions. What is the consequence and the indication? ●
- Loss of the signal, with the lower stop indicated
 - Icing of the probe, with ice temperature indicated
 - Icing of the probe, with the upper stop indicated
- 152.** A TAT indication sits hard against its lower stop and does not move. Which fault fits this picture? ●
- A short circuit in the measuring wire
 - A break in the measuring wire
 - A failure of the probe heating

Answers — Temperature-indicating systems

- 117. a** Temperature is one of the inputs needed for true airspeed. (L1, PDF p. 489)
- 118. c** Air temperature is the indication used to judge whether ice can form. (L1, PDF p. 489)
- 119. b** SAT is the true ambient air temperature, with no ram rise in it. (L1, PDF p. 491)
- 120. b** TAT is ambient temperature plus the heating caused by compression of the air. (L1, PDF p. 491)
- 121. a** Only TAT comes straight off the probe; SAT and the ram rise are worked out from it. (L1, PDF p. 493)
- 122. c** The probe carries two or three temperature dependent PTC resistors. (L1, PDF p. 495)
- 123. a** Air temperature feeds engine performance calculation and the engine limits. (L2, PDF p. 489)

- 124. a The electronic display has no sensing element of its own, so an external OAT sensor is required. (L2, PDF p. 489)
- 125. a The simple system is restricted to small propeller driven aeroplanes and helicopters. (L2, PDF p. 489)
- 126. b Its accuracy runs out at 250 kts, which is why it is limited to slow aircraft. (L2, PDF p. 489)
- 127. b Modern instruments display digitally and do the calculation separately, in the computer, unlike the conventional analogue indicator that carries its switching inside the instrument. (L2, PDF p. 491)
- 128. c SAT cannot be measured directly in flight, so the ADC calculates it from other data. (L2, PDF p. 491)
- 129. b Ram air at the probe produces the ram rise that lifts TAT above SAT. (L2, PDF p. 491)
- 130. a The free airstream over the probe is deflected around the fuselage, which is the built-in RAT error. (L2, PDF p. 491)
- 131. c A flush probe gives ram air temperature; the TAT probe is the one held off the skin. (L2, PDF p. 491)
- 132. c Holding the probe out in the airstream, like a pitot tube, is what lets it sense TAT. (L2, PDF p. 491)
- 133. b Only the measuring resistor sits outside; the rest of the switching is in the instrument. (L2, PDF p. 491)
- 134. b The ADC and FMC use it to work out values that cannot be measured directly. (L2, PDF p. 491)
- 135. a The rise is the ram rise, caused by the air being compressed at the probe. (L2, PDF p. 493)
- 136. b Ram rise is a function of mach number. (L2, PDF p. 493)
- 137. b Slowing the air inside the probe is what creates the ram effect the probe needs to sense TAT. (L2, PDF p. 495)
- 138. c Airflow in flight carries the heat away, and on the ground the heating is normally cut automatically. (L2, PDF p. 495)
- 139. c An aspirated probe uses bleed air to draw outside air through it. (L2, PDF p. 495)
- 140. b An open circuit looks like infinite resistance, which drives the pointer to the upper stop. (L2, PDF p. 495)
- 141. a A short gives zero resistance, which drives the pointer to the lower stop. (L2, PDF p. 495)
- 142. c The elements are PTC resistors, and the characteristic shows resistance rising with temperature. (L2, PDF p. 495)
- 143. c Full ram rise gives the highest value at the TAT probe, RAT loses part of it, and SAT has none. (L3, PDF p. 491)
- 144. b RAT sits between TAT and SAT, so it is still above ambient and the SAT has to be calculated. (L3, PDF p. 491)
- 145. c The simple system is only accurate to 250 kts because ram rise grows with speed, so faster aircraft must have SAT calculated in the ADC. (L3, PDF p. 491)
- 146. a The gap between TAT and SAT is the ram rise, and that grows with mach number. (L3, PDF p. 493)
- 147. c SAT cannot be sensed directly, so the ADC derives it from the measured TAT and the mach number. (L3, PDF p. 493)
- 148. c Page 496 states the basis for temperature measurement is always a resistor bridge, so an open measuring wire gives an infinitely high resistance and the indication goes to the upper stop at +50 C. (L3, PDF p. 495)
- 149. a The slowed air gives the probe the total air temperature, and page 496 states the static air temperature cannot be determined directly but is calculated in the air data computer. (L3, PDF p. 495)
- 150. b The aspirated probe keeps outside air moving through it, so sun and skin temperature on the ground can be ignored. (L3, PDF p. 495)
- 151. b Without heating the probe ices up and then reads the temperature of the ice on it. (L3, PDF p. 495)
- 152. a Only the short circuit, giving 0 ohm, parks the indication at the lower stop. (L3, PDF p. 495)

Analogue air data computer

PDF pages 497–511 • 54 questions

LEVEL 1 — RECALL

- 153.** The abbreviation CADC stands for ●
- a. Corrected Air Data Computer
 - b. Central Air Data Computer
 - c. Combined Air Data Computer
- 154.** The analogue air data computer is built from three modules, which are ●
- a. an altitude module, an airspeed module and a Mach module
 - b. a pitot module, a static module and a temperature module
 - c. an altitude module, a vertical speed module and a temperature module
- 155.** The indicated airspeed depends directly on ●
- a. the dynamic pressure, the difference between total and static pressure
 - b. the static pressure, the difference between total and dynamic pressure
 - c. the total pressure, the sum of static and dynamic pressure
- 156.** MASI is the abbreviation for ●
- a. manual airspeed indicator
 - b. Mach/altitude signal indicator
 - c. Mach/airspeed indicator
- 157.** The true airspeed is the speed of the aircraft measured against ●
- a. the surrounding air
 - b. the standard atmosphere at sea level
 - c. the ground beneath the aircraft
- 158.** When the air data computer has compensated the compressibility error, the result is called ●
- a. equivalent airspeed
 - b. calibrated airspeed
 - c. true airspeed
- 159.** The flag fitted to a vertical speed indicator driven by an air data computer is ●
- a. the STDBY flag
 - b. the OFF flag
 - c. the ALT flag

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 160.** An engineer asks why aircraft manufacturers introduced the central air data computer. What is the reason? ●
- a. It sorted out the muddle of individual pressure instruments
 - b. It made the standby air data instruments unnecessary
 - c. It removed the need for pitot and static ports on the fuselage
- 161.** Among the main uses of the central air data computer is the correction of ●
- a. compass deviation and compass variation
 - b. static source errors and instrument errors
 - c. pitot blockage errors and drain errors

- 162.** The pitot and static supply of the standby air data instruments ●
- a. is shared with the number one air data computer
 - b. is taken from the air data computer in use
 - c. is an additional system of its own
- 163.** One of the main uses of the central air data computer is ●
- a. to measure the deviation of the magnetic compass
 - b. to pass the barometric setting out to the static ports
 - c. to determine values which are not directly measurable
- 164.** A technician traces the vertical speed output of an analogue air data computer. Which module produces it? ●
- a. The airspeed module
 - b. The altitude module
 - c. The Mach module
- 165.** The airspeed module of an analogue air data computer calculates the dynamic pressure from ●
- a. the total pressure and the Mach number
 - b. the static pressure and the total pressure
 - c. the static pressure and the temperature
- 166.** The Mach module of an analogue air data computer takes its information from ●
- a. the altitude module and the airspeed module
 - b. the pitot connection and the static connection
 - c. the aircraft adapter and the pin programming
- 167.** Besides the Mach number, the static source error depends on ●
- a. the length of the pitot lines and the size of the drain holes
 - b. the location of the static ports and the shape of the fuselage
 - c. the position of the temperature probe and the shape of the wing
- 168.** With pin programming, an air data computer receives its aircraft characteristic data as
- a. a coded word on the output data bus
 - b. a ground signal on a specific rack connector pin
 - c. a reference voltage set by the altitude module
- 169.** The reason an air data computer has to be supplied with aircraft characteristic data ●
- a. Because the computer has to be matched to the indicators connected to it
 - b. Because the computer has to correct the compressibility error of the airspeed
 - c. Because the same computer can be used in different aircraft types
- 170.** The purpose of supplying angle of attack, flap position and gear position to some air data computers ●
- a. To make the compressibility error correction more accurate
 - b. To make the static source error correction more accurate
 - c. To make the instrument error correction more accurate
- 171.** In the synchrotel of a servo corrected altimeter, the altitude transmitted by the air data computer is compared with ●
- a. the altitude held by the drum pointer motor
 - b. the altitude set on the baro scale
 - c. the altitude recorded by the pressure capsule

- 172.** If the altitude from the air data computer and the altitude from the capsule do not coincide ●
- an error voltage builds up in the coil of the synchrotel
 - the STDBY flag comes into view at once
 - the baro window is driven back to the standard setting
- 173.** The error voltage in a servo corrected altimeter has fallen to about zero. What does this tell the engineer? ●
- The indication and the altitude transmitted by the air data computer are the same
 - The correction has been switched to standby
 - The pressure capsule has been disconnected from the drum pointer
- 174.** With the switch in STBY, a servo corrected altimeter ●
- works purely mechanically and shows the STDBY flag
 - works purely mechanically and shows the ALT flag
 - works purely electrically and shows the STDBY flag
- 175.** A servo corrected altimeter contains an internal vibrator. What is it there for? ●
- To reduce friction
 - To warn the crew that the correction has dropped out
 - To damp the pointer during rapid climbs
- 176.** In a servo corrected altimeter the barometric correction made with the BARO button is
- purely electrical, through the air data computer
 - purely mechanical
 - a combination of capsule movement and servo drive
- 177.** In a servo corrected altimeter the movement of the pressure capsule reaches the pointer and drum ●
- via the amplifier and the motor
 - via differential gearing
 - via the rotor of the synchrotel
- 178.** The supply voltage to a servo corrected altimeter is lost. Why does the correction stop? ●
- Because the error voltage has to be about zero before the correction can continue
 - Because the ALT flag has to be valid before the correction can continue
 - Because voltage present is one of the requirements for continuous operation of the correction
- 179.** The altitude transmitted to an electrical altimeter by the air data computer is based on
- the elevation of the departure aerodrome
 - the standard of 1013 mbar
 - the pressure set on the baro knob
- 180.** An aircraft makes only small changes of height. What happens to the coarse synchro value in the electrical altimeter? ●
- It is blocked by the OFF flag monitor
 - It is blocked by the HIGH LEVEL detector
 - It is blocked by the LIMITER
- 181.** An aircraft makes a large change of altitude. What happens in the electrical altimeter? ●
- The fine synchro value predominates and the coarse value is suppressed by the LIMITER
 - The coarse synchro value predominates and the fine value is suppressed by the HIGH LEVEL detector
 - The coarse synchro value predominates and the fine value is suppressed by the LIMITER

- 182.** Why is the indicated airspeed treated as a very important parameter for safe flight?
- It gives the real speed of the aircraft at every altitude
 - Lift, drag and control effectiveness all follow the dynamic pressure
 - It is already corrected for instrument and static source error
- 183.** A glass cockpit aircraft displays calibrated airspeed. What has been corrected to obtain it? ●
- The error from the indicator and the error from the compressibility of the air
 - The error from the indicator and the error from the static pressure source
 - The error from the pitot source and the error from the air density
- 184.** CAS is often also taken to mean computed airspeed because ●
- the correction is done by an air data computer
 - the value is computed from the Mach number
 - the value is computed inside the indicator
- 185.** The transmission of the speed value from the air data computer to the indicator ●
- is made as a digital data word in every case
 - is technically the same as for a vertical speed indicator
 - is technically the same as for an altimeter
- 186.** The Mach number is transmitted from the air data computer to the indicator
- in technically the same way as the CAS value
 - as a pneumatic signal from the pitot connection
 - as a discrete that is valid for all parameters
- 187.** A technician is asked where the compressibility error comes from. What is the correct explanation?
- Air compresses at higher speeds and so generates an increase of the total pressure
 - Air density falls with altitude and so the dynamic pressure drops
 - Air compresses at higher altitudes and so the static pressure falls
- 188.** True airspeed and equivalent airspeed are identical ●
- at mean sea level
 - at the speed of sound
 - at the tropopause
- 189.** A student asks why navigation needs the true airspeed rather than the indicated airspeed. What is the correct reply? ●
- Navigation needs the real speed of the aircraft, measured against the ground
 - Navigation needs the speed the structure can stand, measured as dynamic pressure
 - Navigation needs the real speed of the aircraft, measured against the surrounding air
- 190.** The OFF flag of a vertical speed indicator with analogue input monitors ●
- the indicator and the air data computer signal
 - the static source error correction
 - the indicator only

LEVEL 3 — APPLY AND REASON

- 191.** An aircraft is fitted with two air data computers. How are their pressure sources arranged? ●
- Both computers are fed from one common pitot and static system
 - Each computer has its own separated pitot and static system
 - Both computers are fed from the standby pitot and static system

- 192.** A static port feeding one of an aircraft's two air data computers becomes blocked. Why does the other computer still give a correct altitude? ●
- Each of the two air data computers has a separated pitot and static system of its own
 - The two air data computers share one pitot and static system through a crossfeed valve
 - Each of the two air data computers corrects its own static source error from the Mach number
- 193.** The temperature supply to an analogue air data computer is lost. Which outputs can no longer be produced? ●
- Calibrated airspeed and altitude
 - Vertical speed and Mach number
 - True airspeed and static air temperature
- 194.** The Mach module of an analogue air data computer fails. Why is the static source error correction affected as well? ●
- The correction depends primarily on the calibrated airspeed, which comes from that module
 - The correction depends on the static pressure, which is routed through that module
 - The correction depends primarily on the Mach number, which comes from that module
- 195.** The static pressure reaching an analogue air data computer is in error. The effect of this on the Mach indication ●
- The Mach number is wrong as well, because the Mach module works from the temperature input
 - The Mach number is wrong as well, because the Mach module works from the other two modules
 - The Mach number stays correct, because the Mach module has a pressure input of its own
- 196.** A servo corrected altimeter reads several thousand feet away from the altitude sent by the air data computer. Why does the correction not bring it into agreement? ●
- Correction is inhibited while the internal vibrator is running
 - Correction only works above the altitude set on the baro scale
 - Correction is only possible within a few hundred feet
- 197.** In a servo corrected altimeter, the element acted on by the amplifier and motor when an error voltage builds up in the coil of the synchrotel ●
- The pressure capsule, which is corrected until the error voltage approaches zero
 - The altitude sent by the air data computer, which is corrected until the error voltage approaches zero
 - The baro setting, which is corrected until the error voltage approaches zero
- 198.** The STDBY flag is in view on a servo corrected altimeter. What does this tell the engineer about how the instrument is working? ●
- The correction is active and the instrument is following the air data computer
 - The correction is deactivated and the instrument is working from the synchrotel alone
 - The correction is deactivated and the instrument is working purely mechanically
- 199.** In an electrical altimeter, the element that blocks the coarse synchro value on small changes of height and the element that suppresses the fine value on large changes ●
- The LIMITER blocks the coarse value and the HIGH LEVEL detector suppresses the fine value
 - The HIGH LEVEL detector blocks the coarse value and the baro correction suppresses the fine value
 - The HIGH LEVEL detector blocks the coarse value and the LIMITER suppresses the fine value
- 200.** An aircraft climbs at a constant indicated airspeed. Why does the IAS no longer give the real speed of the aircraft?
- The indicator is calibrated to the standard air pressure and the real pressure is not taken into account
 - The indicator is calibrated to the standard air density and the real density is not taken into account
 - The dynamic pressure is no longer the difference between total and static pressure

- 201.** An air data computer stops working altogether. What is seen on the Mach/airspeed indicator? ●
- No flag at all, because the indicator holds the last values received
 - A red flag for each value, because the valid discretes are no longer transmitted
 - A red flag for the Mach value only, because CAS is taken from a capsule
- 202.** A red flag shows against one parameter on a Mach/airspeed indicator while the other parameter reads normally. What should the engineer conclude? ●
- The fault must lie in the air data computer
 - The fault must lie in the pitot and static system
 - The fault may lie either in the air data computer or in the indicator
- 203.** A red flag is shown against both the Mach number and the airspeed on a Mach/airspeed indicator. Which two faults could give this? ●
- An internal fault in the indicator, or a loss of the indicator lighting supply
 - An air data computer fault that withholds the valid discretes, or a blockage in the static line
 - An air data computer fault that withholds the valid discretes, or an internal fault in the indicator
- 204.** How does the air data computer obtain true airspeed from equivalent airspeed? ●
- It replaces the standard pressure used before with the real pressure
 - It replaces the standard density used before with the real density
 - It adds the static source error correction to the equivalent airspeed
- 205.** An aircraft climbs while holding a constant equivalent airspeed. What happens to the true airspeed? ●
- It stays the same, because the two values are identical at every altitude
 - It increases, because the true airspeed rises with increasing altitude
 - It decreases, because the true airspeed falls with increasing altitude
- 206.** The correction made between calibrated airspeed and equivalent airspeed, and the one made between equivalent airspeed and true airspeed ●
- The standard density is replaced with the real density, then the compressibility error is compensated
 - The compressibility error is compensated, then the standard density is replaced with the real density
 - The compressibility error is compensated, then the static source error is corrected

Answers — Analogue air data computer

- 153. b** The manual expands CADC as Central Air Data Computer, also called ADC in short. (L1, PDF p. 497)
- 154. a** The manual names altitude, airspeed and Mach as the three modules. (L1, PDF p. 499)
- 155. a** IAS is a direct function of dynamic pressure, which is total minus static pressure. (L1, PDF p. 505)
- 156. c** The instrument combines the Mach and airspeed indications in one display. (L1, PDF p. 507)
- 157. a** TAS is the real speed of the aircraft relative to the air around it. (L1, PDF p. 509)
- 158. a** Compensating compressibility turns CAS into EAS. (L1, PDF p. 509)
- 159. b** The vertical speed indicator with analogue input carries an OFF flag. (L1, PDF p. 511)
- 160. a** Conventional air data conversion meant installing many separate aneroid and differential pressure instruments; the CADC replaced that muddle. (L2, PDF p. 497)
- 161. b** Correcting static source error and instrument error is listed as a main task of the CADC. (L2, PDF p. 497)
- 162. c** The standby air data instruments have their own pitot and static system in addition to those of the computers. (L2, PDF p. 497)
- 163. c** Deriving values that cannot be measured directly is listed as one of the CADC main tasks. (L2, PDF p. 497)
- 164. b** The altitude module converts static pressure into both the altitude and the vertical speed signal. (L2, PDF p. 499)
- 165. b** Dynamic pressure is derived in the airspeed module from static and total pressure, giving CAS. (L2, PDF p. 499)

- 166. a** The Mach module works from the outputs of the other two modules rather than from the pressures directly. (L2, PDF p. 499)
- 167. b** Static port location and fuselage shape are the further factors given for the static source error. (L2, PDF p. 499)
- 168. b** Pin programming grounds a particular rack connector pin to tell the computer which aircraft type it is in. (L2, PDF p. 499)
- 169. c** One computer can be fitted to different aircraft types, so it must be told the characteristics of the aircraft it is installed in. (L2, PDF p. 499)
- 170. b** These extra inputs are used to sharpen the static source error correction. (L2, PDF p. 499)
- 171. c** The synchrotel compares the ADC altitude against the capsule altitude. (L2, PDF p. 501)
- 172. a** A difference between the two altitudes produces an error voltage in the synchrotel coil. (L2, PDF p. 501)
- 173. a** Zero error voltage is the state in which indication and transmitted altitude agree. (L2, PDF p. 501)
- 174. a** Deactivating the correction leaves a purely mechanical instrument with the STDBY flag in view. (L2, PDF p. 501)
- 175. a** The vibrator is fitted to reduce friction inside the mechanism. (L2, PDF p. 501)
- 176. b** The baro setting acts mechanically on the instrument, not through the computer. (L2, PDF p. 501)
- 177. b** The capsule acts on the pointer and drum through differential gearing. (L2, PDF p. 501)
- 178. c** Voltage present is listed as one of the requirements for the correction to keep running, so losing the supply stops it. (L2, PDF p. 501)
- 179. b** The computer sends altitude referenced to the standard setting; the baro correction is applied afterwards in the indicator. (L2, PDF p. 503)
- 180. b** For small height changes the fine value alone is used and the HIGH LEVEL detector blocks the coarse value. (L2, PDF p. 503)
- 181. c** Large changes let the coarse value take over, with the LIMITER suppressing the fine value. (L2, PDF p. 503)
- 182. b** IAS matters because the aerodynamic effects and control effectiveness follow dynamic pressure. (L2, PDF p. 505)
- 183. b** CAS is IAS corrected for instrument error and static source error. (L2, PDF p. 505)
- 184. a** The computer performs the correction, which is why the second reading of the abbreviation is used. (L2, PDF p. 505)
- 185. c** The speed value is sent to its indicator in technically the same way as the altitude value is sent to an altimeter. (L2, PDF p. 505)
- 186. a** Mach uses the same transmission technique as the calibrated airspeed value. (L2, PDF p. 507)
- 187. a** The speed formula assumes an incompressible medium, and the compression of the air at speed raises the total pressure. (L2, PDF p. 509)
- 188. a** Only at mean sea level does the real density equal the standard density used for EAS. (L2, PDF p. 509)
- 189. c** Navigation needs the real speed of the aircraft, and that is the true airspeed, measured against the surrounding air. (L2, PDF p. 509)
- 190. a** The flag covers both the indicator itself and the signal arriving from the computer. (L2, PDF p. 511)
- 191. b** Each of the two computers is given a separate pitot and static system of its own. (L3, PDF p. 497)
- 192. a** Because the two computers have separate pitot and static systems, a blockage in one system cannot reach the other computer. (L3, PDF p. 497)
- 193. c** TAS and SAT are the outputs that depend on the temperature being supplied to the Mach module. (L3, PDF p. 499)
- 194. c** Static source error depends first of all on Mach number, and that value is supplied by the Mach module. (L3, PDF p. 499)
- 195. b** The Mach module takes its information from the altitude and airspeed modules, both of which use the static pressure, so an error in that pressure carries into the Mach number. (L3, PDF p. 499)
- 196. c** Continuous correction requires the error voltage to be about zero, which limits it to a few hundred feet. (L3, PDF p. 501)
- 197. a** The error voltage is amplified and drives the motor, which corrects the pressure capsule until the error voltage is about zero; the baro setting is a purely mechanical correction made with the BARO button. (L3, PDF p. 501)
- 198. c** The STDBY flag goes with the correction being deactivated, and in that condition the instrument works purely mechanically. (L3, PDF p. 501)
- 199. c** On small changes the HIGH LEVEL detector blocks the coarse value, and on large changes the LIMITER suppresses the fine value. (L3, PDF p. 503)

- 200. b** Calibration to standard density means IAS is only the real speed at mean sea level under standard conditions. (L3, PDF p. 505)
- 201. b** Loss of the valid discrete for a value raises the red flag for that value. (L3, PDF p. 507)
- 202. c** The indicator raises the same flag for its own internal faults, so the flag alone does not isolate the computer. (L3, PDF p. 507)
- 203. c** A flag follows either a missing valid discrete from the air data computer or an internal fault found by the indicator itself. (L3, PDF p. 507)
- 204. b** The step from EAS to TAS is the substitution of real air density for standard density. (L3, PDF p. 509)
- 205. b** The manual states that TAS increases with increasing altitude for a given EAS. (L3, PDF p. 509)
- 206. b** Compensating the compressibility error gives the equivalent airspeed, and replacing the standard density with the real density then gives the true airspeed. (L3, PDF p. 509)

Hybrid and digital air data computers

PDF pages 513–525 · 21 questions

LEVEL 1 — RECALL

- 207.** The analogue output data of a hybrid air data computer is provided in the format ●
- ARINC 429
 - ARINC 629
 - ARINC 565
- 208.** The pressure transducers of a digital air data computer are used to ●
- hold the input pressures at a reference value
 - convert the input pressures into a mechanical movement
 - convert the input pressures into electrical signals
- 209.** The data format used by a digital air data computer is usually ●
- ARINC 565
 - ARINC 429
 - ARINC 573

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 210.** A hybrid air data computer is described as hybrid because ●
- it provides output data in analogue form as well as digitally
 - it can be supplied with either AC or DC power
 - it uses both pressure capsules and pressure transducers
- 211.** An analogue air data computer is unserviceable and a DADC is offered as a replacement. What applies? ●
- The DADC may not be installed in place of an analogue computer
 - The DADC may only be installed if both computers are changed together
 - The DADC may be installed in place of the analogue computer
- 212.** Compared with an analogue computer, the internal arrangement of a hybrid air data computer
- has an additional module for true airspeed
 - keeps the three modules but adds a digital output stage
 - has no division into altitude, airspeed and Mach modes
- 213.** Two types of pressure transducer used in digital air data computers are ●
- the resistive bridge and the differential bellows
 - the resistive bridge and the vibrating cylinder
 - the aneroid capsule and the vibrating cylinder
- 214.** A technician asks how the digital computer of a DADC produces its many outputs. What is correct?
- It calculates them one after the other and refreshes the data in a predetermined cycle
 - It calculates a value only when an indicator asks for it
 - It calculates them all at the same instant and refreshes them continuously
- 215.** The computed values of a digital air data computer reach the instruments and other users ●
- over one single analogue data bus
 - over separate synchro lines for each value
 - over several digital data busses

- 216.** A digital air data computer normally learns which aircraft type it is fitted to from ●
- a pin programming
 - the angle of attack sensor
 - the temperature probe
- 217.** An inertial reference unit mixes its own accelerometer signal with the air data computer signal before sending vertical speed. Why? ●
- the accelerometer signal alone cannot be converted to a digital output
 - the quick accelerometer signal and the slow air data signal together give a smooth indication
 - the air data signal corrects the accelerometer for position error
- 218.** The disadvantage of a vertical speed signal taken from an air data computer ●
- it shows the vertical speed with a delay
 - it shows the vertical speed only above a set altitude
 - it shows the vertical speed without any delay
- 219.** The inertial reference unit is unable to calculate vertical speed. What appears on the indicator? ●
- A red AS flag
 - A yellow IVS flag
 - A yellow ALT flag
- 220.** The second back up fitted to some vertical speed indicators
- the offside air data computer
 - the onside inertial reference unit
 - the offside standby altimeter

LEVEL 3 — APPLY AND REASON

- 221.** An aircraft has analogue electrical indicators and also users on a digital data bus. Which air data computer can serve both, and how? ●
- A hybrid computer, as it provides output data in analogue form only and feeds the bus users through the indicators
 - An analogue computer, as it provides output data in analogue form (ARINC 565) as well as digitally (ARINC 429)
 - A hybrid computer, as it provides output data in analogue form (ARINC 565) as well as digitally (ARINC 429)
- 222.** A digital air data computer has lost its angle of attack, flap and gear position inputs. Which of its functions is degraded? ●
- The transmission of the computed values on the output busses
 - The conversion of the input pressures into electrical signals
 - The data corrections, such as the static source error correction
- 223.** The inputs a digital air data computer takes from the aircraft sensing elements, and the form in which its computed values leave it ●
- Pitot pressure, static pressure and temperature in, digital data on several data busses out
 - Pitot pressure, static pressure and temperature in, analogue signals on separate wires out
 - Pitot pressure, static pressure and dynamic pressure in, digital data on several data busses out

- 224.** The barometric setting is changed on an altimeter with digital input. Which way does that signal travel?
- In analogue form from the indicator to the DADC
 - In digital form from the indicator to the DADC
 - In analogue form from the DADC to the indicator
- 225.** The air data computers have failed, yet the digital vertical speed indicator still gives a usable indication. Why? ●
- It normally works from the vertical speed output of the air data computer
 - It normally works from the inertial vertical speed signal of the inertial reference unit
 - It normally works from the static pressure rate measured in the indicator itself
- 226.** In flight the inertial vertical speed signal from the inertial reference unit is lost. What happens at the vertical speed indicator? ●
- The vertical speed from the air data computer is used automatically
 - The indicator freezes at the last valid vertical speed
 - The indicator blanks until another source is selected by hand
- 227.** A pilot's air data selector is moved out of its normal position. Which indications change over?
- That pilot's vertical speed indicator goes to the other digital air data computer
 - That pilot's standby instruments go to the other digital air data computer
 - That pilot's airspeed indicator and altimeter go to the other digital air data computer

Answers — Hybrid and digital air data computers

- 207. c** ARINC 565 is the analogue format, ARINC 429 the digital one. (L1, PDF p. 513)
- 208. c** The transducer turns pitot and static pressure into an electrical quantity. (L1, PDF p. 515)
- 209. b** ARINC 429 is given as the usual format for the digital computer output. (L1, PDF p. 515)
- 210. a** The mixture of analogue and digital outputs is what makes the computer hybrid. (L2, PDF p. 513)
- 211. c** The replacement is allowed in that direction only, because the DADC also provides analogue outputs. (L2, PDF p. 513)
- 212. c** The hybrid computer drops the modular split used in the analogue computer. (L2, PDF p. 513)
- 213. b** The manual names the resistive bridge transducer and the vibrating cylinder. (L2, PDF p. 515)
- 214. a** Outputs are computed in sequence and refreshed on a fixed cycle. (L2, PDF p. 515)
- 215. c** Distribution is by several digital data busses. (L2, PDF p. 515)
- 216. a** Aircraft type information comes from pin programming; the other inputs serve the correction itself. (L2, PDF p. 515)
- 217. b** Mixing the fast accelerometer signal with the slow air data signal produces a smooth vertical speed indication. (L2, PDF p. 523)
- 218. a** The air data derived signal lags, which is why the inertial signal is preferred. (L2, PDF p. 523)
- 219. b** Loss of the inertial vertical speed calculation is annunciated by the yellow IVS flag. (L2, PDF p. 523)
- 220. a** Beyond the onside air data computer, some indicators can also use the offside air data computer. (L2, PDF p. 523)
- 221. c** The hybrid computer is so called because it puts out both analogue ARINC 565 data and digital ARINC 429 data, so it can feed both kinds of user. (L3, PDF p. 513)
- 222. c** Those inputs are used for the data corrections, so only the correction accuracy suffers. (L3, PDF p. 515)
- 223. a** The computer is fed with pitot pressure, static pressure and temperature, and sends its computed values out digitally on several data busses; dynamic pressure is not sensed but worked out from the two pressures. (L3, PDF p. 515)
- 224. a** The baro value runs back the other way, from the indicator to the computer, and in analogue form. (L3, PDF p. 519)
- 225. b** The normal source is the IRU, so the indication survives the loss of the air data computers. (L3, PDF p. 523)
- 226. a** The changeover to air data vertical speed is automatic, so the indication continues. (L3, PDF p. 523)

- 227. c** The air data selector transfers the air data indicators, airspeed and altitude, to the other air data computer; the vertical speed indicator is fed from an inertial source. (L3, PDF p. 525)

Digital indicators and source selection

PDF pages 527–539 • 25 questions

LEVEL 1 — RECALL

- 228.** Installing a glass cockpit in an aircraft with digital air data computers ●
- replaces the digital air data computers
 - replaces the individual indicators
 - replaces the pitot probes and static ports
- 229.** The primary flight display is a display which shows ●
- the true airspeed only
 - the engine parameters
 - most of the air data
- 230.** EFIS means ●
- Electronic Flight Display System
 - Electronic Flight Instrument Switching
 - Electrical Flight Director System
- 231.** The air data indication shown on the navigation display ●
- barometric altitude
 - computed airspeed
 - true airspeed
- 232.** In a system built around air data inertial reference units, the air data modules ●
- amplify the air data computer output for the indicators
 - convert the total air temperature instead of air data computers
 - convert the pitot and static pressure instead of air data computers
- 233.** ADIRU is short for ●
- air data integrated reference unit
 - air data inertial reference unit
 - airspeed and altitude data reference unit
- 234.** ADM means ●
- Air Data Monitor
 - Altitude Data Module
 - Air Data Module
- 235.** The power supply for an air data module comes from ●
- the related air data inertial reference unit
 - the essential AC busbar direct
 - the related display management computer

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 236.** An aircraft has a glass cockpit. Where does the crew read airspeed, altitude and vertical speed? ●
- On the navigation display
 - On the primary flight display
 - On the display management computer

- 237.** The equipment an aircraft must have before a glass cockpit can be installed in place of its indicators ●
- Analogue air data computers
 - Digital air data computers
 - An air data module at every port
- 238.** On many aircraft the baro select knob can be pulled or pushed. What does this do?
- It sets the field elevation as the altitude reference
 - It sets standard baro correction immediately
 - It changes the altitude display to metric
- 239.** In a glass cockpit, the separate indicators replaced by the navigation display ●
- the vertical speed indicators
 - the true airspeed indicators
 - the standby instruments
- 240.** An air data module is opened up for inspection. What sort of pressure transducer is found inside? ●
- a simpler analogue type than those used in the digital air data computer
 - aneroid capsules of a type not used in the digital air data computer
 - the same type as are used in the digital air data computer
- 241.** The reason the air data computer can be integrated with the inertial reference unit
- the computer consists of the digital parts only and so requires little space
 - the air data is now computed by the display management computers
 - the inertial reference unit no longer needs its own accelerometers
- 242.** Air data modules are located near the static port or pitot probe and there they ●
- convert the air pressure into a proportional analogue voltage
 - convert the air pressure directly into a digital data word
 - amplify the air pressure for transmission along the tubes
- 243.** In the latest generation of air data module, the static port or pitot probe is united with the module so that ●
- there is only a short flexible hose between them
 - there is no tube between them
 - there is no electrical connector between them
- 244.** An aircraft fitted with air data modules is lighter and cheaper to maintain than one with long pneumatic runs. Why? ●
- the pressure transducers are simpler than those in a digital air data computer
 - the modules need no electrical power supply
 - the data is sent to the computers via a digital data bus

LEVEL 3 — APPLY AND REASON

- 245.** The altitude shown on the primary flight displays disagrees with the standby instrument. Which settings should be compared? ●
- The baro correction on the EFIS control panel and the BARO setting on the standby instrument
 - The baro correction on the EFIS control panel and the air data source selection
 - The BARO setting on the standby instrument and the selected altitude on the mode control panel

- 246.** A pilot turns the baro knob on the EFIS control panel. Where is the result of that selection seen? ●
- As a corrected true airspeed on the primary flight display
 - As the selected baro correction on the navigation display
 - As the selected baro correction on the primary flight display
- 247.** A student looks for the Mach number on the navigation display and cannot find it. Why not? ●
- True airspeed is the only air data indication on that display
 - Mach number is only shown above the Mach changeover altitude
 - Mach number is only shown when the autopilot is engaged
- 248.** Both navigation displays of a glass cockpit aircraft fail while the primary flight displays stay serviceable. Which air data value is no longer indicated? ●
- The calibrated airspeed
 - The barometric altitude
 - The true airspeed
- 249.** An aircraft has air data modules fitted at its probes and ports. What does its air data computing section no longer need? ●
- An analogue pressure sensing section of its own
 - A connection to the inertial reference unit
 - A digital output to the display management computers
- 250.** Pitot pressure is converted into a digital data word at the probe. Which path does that data then follow? ●
- To the standby instruments, and from there on to an air data inertial reference unit
 - To the display management computers, and from there on to an air data inertial reference unit
 - To an air data inertial reference unit, and from there on to the user systems
- 251.** An aircraft uses air data modules at its probes and ports. Which equipment must still be supplied by pneumatic tubes? ●
- The standby instruments
 - The air data inertial reference units
 - The primary flight displays
- 252.** An air data inertial reference unit is removed for repair. What is the consequence for the air data module associated with it? ●
- The module is powered instead by the display management computer
 - The module continues to work on its own internal battery
 - The module loses its power supply

Answers — Digital indicators and source selection

- 228. b** The glass cockpit takes the place of the separate indicators, not of the computers or the sensing system. (L1, PDF p. 527)
- 229. c** The PFD is where most of the air data is presented. (L1, PDF p. 527)
- 230. a** The manual expands EFIS as Electronic Flight Display System. (L1, PDF p. 529)
- 231. c** True airspeed is the one air data value presented on the navigation display. (L1, PDF p. 531)
- 232. c** The pressure conversion that an air data computer used to do is done by the air data modules. (L1, PDF p. 533)
- 233. b** ADIRU is the name of the combined air data and inertial reference box. (L1, PDF p. 533)
- 234. c** The abbreviation is expanded in the manual as Air Data Module. (L1, PDF p. 539)
- 235. a** Each module is powered by the ADIRU it serves. (L1, PDF p. 539)
- 236. b** The PFD carries most of the air data, including airspeed, altitude and vertical speed. (L2, PDF p. 527)

- 237. b** A glass cockpit replacing the indicators is possible in aircraft fitted with digital air data computers. (L2, PDF p. 527)
- 238. b** Pulling or pushing the knob is a shortcut to the standard setting. (L2, PDF p. 529)
- 239. b** The ND takes over the function of the TAS indicators. (L2, PDF p. 531)
- 240. c** The transducer itself is unchanged; only its location and the form of its output have moved. (L2, PDF p. 533)
- 241. a** Once the pressure sensing has moved out to the modules, the computer is small enough to share a box with the IRU. (L2, PDF p. 533)
- 242. b** Conversion to a digital data word happens at the module, right by the sensing point. (L2, PDF p. 539)
- 243. b** Uniting the sensing point with the module removes the pneumatic tube altogether. (L2, PDF p. 539)
- 244. c** Wiring a data bus replaces long runs of pneumatic tubing, which is where the weight and maintenance saving comes from. (L2, PDF p. 539)
- 245. a** The displays and the standby instrument take their barometric reference from separate controls, so a disagreement points to the two settings being different. (L3, PDF p. 529)
- 246. c** The setting made on the EFIS control panel is displayed as the selected baro correction on the PFD, which carries the air data. (L3, PDF p. 529)
- 247. a** The ND carries TAS alone; the rest of the air data, Mach included, is on the primary flight display. (L3, PDF p. 531)
- 248. c** The navigation display carries the true airspeed and no other air data, and it has replaced the separate TAS indicators, so losing it loses the TAS indication. (L3, PDF p. 531)
- 249. a** With the modules doing the pressure conversion the computer keeps only digital parts, which is what makes it small enough to combine with the IRU. (L3, PDF p. 533)
- 250. c** The module feeds the ADIRU, which is the computer that produces the air data outputs for the displays and other systems. (L3, PDF p. 539)
- 251. a** Everything else is fed by data bus; the standby instruments keep their pneumatic supply. (L3, PDF p. 539)
- 252. c** Because the module takes its power from the related ADIRU, removing that unit leaves the module unpowered. (L3, PDF p. 539)

ADIRU, air data modules and standby instruments

PDF pages 541–547 · 30 questions

LEVEL 1 — RECALL

- 253.** The three tasks united in one box by a GNADIRU ●
- a. inertial reference unit, display management computer and GPS receiver
 - b. digital air data computer, inertial reference unit and GPS receiver
 - c. digital air data computer, air data module and GPS receiver
- 254.** The main signal generators of the inertial reference section ●
- a. the acceleration sensors and the GPS antenna
 - b. the ring laser gyro and the acceleration sensors
 - c. the ring laser gyro and the pressure transducers

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 255.** The data supplied by the air data components of a GNADIRU ●
- a. airspeed, Mach, angle of attack, temperature and barometric altitude
 - b. attitude, heading, vertical acceleration and body rates
 - c. present position, satellite status and receiver clock error
- 256.** Total and static pressure reach a GNADIRU ●
- a. through the display management computers
 - b. from air data modules on data buses
 - c. through pneumatic tubes direct to the unit
- 257.** The data supplied by the inertial reference component of a GNADIRU ●
- a. total pressure and static pressure converted at the probes
 - b. attitude, flight path vector, speed above ground and position
 - c. present position and other parameters from the satellites
- 258.** The advantage of the ring laser gyro over an older mechanical gyroscope ●
- a. it needs no electrical power supply
 - b. it has no moving parts at all
 - c. it needs no alignment before flight
- 259.** The GPS receiver inside a GNADIRU calculates present position and transmits it ●
- a. to the air data modules on data buses
 - b. to the standby instruments on discrete lines
 - c. to the flight manager system on data buses
- 260.** The difference between the first laser gyros and the latest ones ●
- a. The first were based on glass fibre technology, the latest are equipped with prisms
 - b. The first had moving parts, the latest are based on glass fibre technology
 - c. The first were equipped with prisms, the latest are based on glass fibre technology
- 261.** The way the air data section of a GNADIRU works ●
- a. It operates like a digital air data computer
 - b. It operates like an air data module
 - c. It operates like an analogue air data computer

- 262.** At low altitudes the maximum operating velocity depends on ●
- the speed of sound at that altitude
 - the maximum allowed Mach number
 - the dynamic pressure acting on the structure
- 263.** As altitude increases, VMO ●
- decreases steadily as the altitude rises
 - increases sharply as the altitude rises
 - is either constant or increases slightly
- 264.** VMO increases slightly with increasing altitude when ●
- the position error of the static system is taken into account
 - the fall in the speed of sound is taken into account
 - the compressibility error of the indicated airspeed is taken into account
- 265.** An aircraft climbs at a constant indicated airspeed and the Mach limit comes down to meet it. Why does MMO fall with altitude? ●
- the dynamic pressure increases
 - the compressibility error decreases
 - the speed of sound decreases
- 266.** The digital air data computer triggers the overspeed warning discrete
- as soon as the aircraft climbs above the VMO to MMO changeover altitude
 - as soon as the indicated airspeed exceeds VMO or MMO by a set margin
 - as soon as the indicated airspeed is as big as VMO or MMO
- 267.** The form taken by the overspeed warning in the cockpit ●
- a red flag appearing in the airspeed indicator
 - a red light or message with an aural warning sound
 - an amber caution message with no aural sound
- 268.** A pilot wants to read the Mach number on the ISIS. Is that available without the optional display functions?
- Yes, Mach number is in the basic display list
 - No, it appears only with the optional functions
 - No, the ISIS shows true airspeed in its place

LEVEL 3 — APPLY AND REASON

- 269.** A GNADIRU is fitted but the aircraft has no air data computer as a separate box. How does the unit obtain its pressures? ●
- From air data modules placed as near as possible to the pitot and static sensors
 - From the inertial reference section, which derives them from acceleration
 - From pneumatic tubes run from the probes to the unit in the avionics bay
- 270.** The antenna connection to the GPS part of a GNADIRU is broken. Which function of the box is lost, and which data stops reaching the flight management system? ●
- The GPS receiver function, and the present position it calculates
 - The inertial reference function, and the present position it calculates
 - The GPS receiver function, and the barometric altitude it calculates

- 271.** The barometric altitude from a GNADIRU is in error while its attitude and speed above ground are correct. The section at fault ●
- The air data section, which supplies the attitude and speed above ground
 - The air data section, which supplies the barometric altitude
 - The inertial reference section, which supplies the barometric altitude
- 272.** The high speed limit marking on a jet aircraft airspeed display moves as the aircraft climbs. What accounts for this?
- The marking is driven by the selected airspeed from the autopilot mode control panel
 - The indicator processor moves the marking by a fixed amount for each altitude band
 - The digital air data computer sends an overspeed limit that varies with altitude
- 273.** An aircraft is cruising at high altitude. Which of the two speed limits is the governing one there? ●
- VMO, because the compressibility error of the indicated airspeed grows with altitude
 - MMO, because at higher altitudes the limit depends on the maximum allowed Mach number
 - VMO, because at higher altitudes the dynamic pressure on the structure is greatest
- 274.** A digital air data computer fails completely. What is lost in addition to its air data indications?
- Only the maximum allowable airspeed pointer, because the warning comes from the indicator
 - Nothing, because the warning system calculates the overspeed limit itself
 - The overspeed warning, because the limit and the warning discrete come from that computer
- 275.** An integrated standby instrument system fails completely. Which standby indications are lost? ●
- Standby altitude, standby airspeed and standby vertical speed
 - Standby altitude, standby attitude and standby airspeed
 - Standby altitude, standby attitude and standby heading
- 276.** Altitude, airspeed and attitude all disappear from an ISIS indicator although its pressure sensors and gyrometers are serviceable. Which module is suspect?
- The pressure module
 - The computation module
 - The inertial module
- 277.** The static line to an ISIS indicator becomes blocked. Which indications are affected?
- Attitude is affected, while altitude and airspeed continue from the computation module
 - All indications are lost, because every parameter is derived from the pressure module
 - Altitude and airspeed are affected, while attitude continues from the inertial module
- 278.** A technician has to set a QNH on an ISIS indicator. Which control is used, and in what unit is the reference shown as standard?
- The BARO rotary knob, with the reference pressure shown in inches of mercury
 - The BARO rotary knob, with the reference pressure shown in hectopascals
 - The LS pushbutton switch, with the reference pressure shown in hectopascals
- 279.** The magnetic heading indication on an ISIS indicator is missing while its attitude, altitude and airspeed are normal. Which input is suspect?
- The discrete input from the attitude and heading selector switch
 - The digital input from the ILS or multi mode receiver
 - The digital input from the inertial reference

- 280.** The LS function cannot be engaged on an ISIS indicator although the pushbutton switch is serviceable. Which input should be checked?
- The digital input from the inertial reference
 - The digital input from the ILS or multi mode receiver
 - The digital output to the central maintenance system
- 281.** All digital inputs to the ISIS indicator are lost. Which indications can the unit still produce?
- Only altitude, because the pressure module is the only module left working
 - None, because every displayed parameter arrives as a digital input
 - Attitude, altitude and airspeed, from its own inertial and pressure modules
- 282.** Air data modules digitise the pressures everywhere else on the aircraft, yet the ISIS indicator still has pneumatic connectors. Why?
- Because the standby instruments still have to be supplied by tubes
 - Because the ISIS pressure module needs a reference for its gyrometers
 - Because the air data modules cannot work with total pressure

Answers — ADIRU, air data modules and standby instruments

- 253. b** The GNADIRU is an ADIRU with a GPS receiver added. (L1, PDF p. 541)
- 254. b** Gyro and accelerometers together generate the inertial signals. (L1, PDF p. 541)
- 255. a** These are the air data outputs; attitude and position come from the other two sections. (L2, PDF p. 541)
- 256. b** The modules sense the pressures and pass the data on digitally. (L2, PDF p. 541)
- 257. b** These are the inertial outputs of the unit. (L2, PDF p. 541)
- 258. b** The absence of moving parts is what makes it robust and light. (L2, PDF p. 541)
- 259. c** The GPS output is passed on digitally to the flight manager system. (L2, PDF p. 541)
- 260. c** The first laser gyros used prisms and the latest ones use glass fibre technology. (L2, PDF p. 541)
- 261. a** The air data components of the GNADIRU operate like a DADC, a digital air data computer. (L2, PDF p. 541)
- 262. c** VMO is a structural limit set by dynamic pressure. (L2, PDF p. 543)
- 263. c** VMO holds steady or rises a little with height. (L2, PDF p. 543)
- 264. c** Allowing for compressibility error in the indicated airspeed is what makes the limit rise with height. (L2, PDF p. 543)
- 265. c** A falling speed of sound lowers the airspeed corresponding to the maximum Mach number. (L2, PDF p. 543)
- 266. c** The discrete is produced on reaching the limit, with no extra margin. (L2, PDF p. 543)
- 267. b** Overspeed is a warning level alert: red visual plus an aural sound. (L2, PDF p. 543)
- 268. a** Mach number is in the standard display list, whereas true airspeed and vertical speed are not. (L2, PDF p. 545)
- 269. a** The air data section works like a digital air data computer but is fed digitally by modules mounted close to the probes and ports. (L3, PDF p. 541)
- 270. a** The GPS receiver needs the antenna to receive the satellite signals, so without it the present position it calculates and sends to the flight management system is lost. (L3, PDF p. 541)
- 271. b** Barometric altitude comes from the air data part of the box, while attitude and speed above ground come from the inertial reference part, which is working normally. (L3, PDF p. 541)
- 272. c** The limit itself is computed in the air data computer and changes with altitude, and the indicator simply displays the value it is sent. (L3, PDF p. 543)
- 273. b** VMO is the structural limit that rules low down; higher up the Mach limit becomes the lower of the two. (L3, PDF p. 543)
- 274. c** Both the limit and the discrete that triggers the warning originate in the air data computer. (L3, PDF p. 543)
- 275. b** One unit does the work of the three separate standby instruments, so all three indications go together. (L3, PDF p. 545)
- 276. b** All three parameters are worked out, and the picture generated, in the computation module, so a single fault there takes out everything at once. (L3, PDF p. 545)

- 277. c** Attitude comes from the gyrometers and accelerometers of the inertial module, so it is independent of the pressure supply. (L3, PDF p. 545)
- 278. b** The rotary knob sets the pressure and the standard display of the reference is in hectopascals, inches of mercury being an option. (L3, PDF p. 545)
- 279. c** Heading is the only thing that digital input provides, and the parameters the unit computes for itself are unaffected. (L3, PDF p. 547)
- 280. b** The LS display depends on the digital input from the ILS or multi mode receiver. (L3, PDF p. 547)
- 281. c** The ISIS senses pressure and motion itself, so its main parameters do not depend on the digital inputs. (L3, PDF p. 547)
- 282. a** ISIS is the standby instrument, and standby instruments keep their own pneumatic supply so they stay independent of the digital system. (L3, PDF p. 547)

Angle of attack sensing and stall warning

PDF pages 551–562 · 44 questions

LEVEL 1 — RECALL

- 283.** The angle that an angle of attack sensing system determines ●
- the angle between the air stream and the vertical axis of the aircraft
 - the angle between the wing chord and the horizontal
 - the angle between the air stream and the wing chord
- 284.** The lift generated by a wing depends mainly on ●
- the wing geometry, the altitude and the angle of attack
 - the airspeed, the air temperature and the angle of attack
 - the wing geometry, the airspeed and the angle of attack
- 285.** Alpha max is the angle of attack at which ●
- the airflow cannot follow the upper surface of the wing and separation occurs
 - the greatest airspeed is obtained for a given wing geometry
 - the slats and flaps must be extended to keep the lift constant
- 286.** The angle between the air stream and the wing chord ●
- The angle of attack
 - The dihedral angle
 - The angle of incidence
- 287.** The condition present at the beginning of the loss of airflow ●
- Stall speed
 - Initial buffet
 - Minimum manoeuvre speed
- 288.** The name given to the digital stall warning system ●
- Stall warning monitor card system
 - Stall warning management computer system
 - Stall warning controller system
- 289.** The three parts that make up the SWMCS ●
- Stall warning management computer, flap position transmitter and stick shaker
 - Stall warning management computer, master monitor card and stick shaker
 - Master monitor card, angle of attack sensor and stick shaker

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 290.** The relationship between angle of attack and the lift acting on the wings ●
- the greater the angle of attack, the greater the lift
 - the lift is independent of the angle of attack
 - the greater the angle of attack, the smaller the lift
- 291.** If the individual maximum angle of attack of a wing profile is exceeded ●
- the airflow over the surface breaks down
 - the lift continues to increase steadily
 - the angle between the air stream and the wing chord becomes zero

- 292.** A lift transducer determines the angle of attack by ●
- sensing changes in direction and intensity of the stagnation point below the forward wing surface
 - comparing the pressures at two rows of holes in a tube lying crosswise to the air stream
 - sensing the angle of the air stream with a vane on the side of the fuselage
- 293.** An angle of attack sensor is a tube lying crosswise to the air stream, pierced by two rows of holes. How does it sense the angle? ●
- by sensing the movement of the stagnation point inside the tube
 - by measuring the total pressure at the forward end of the tube
 - by comparing the pressures at the two rows
- 294.** The wind vane type of angle of attack sensor is made force-free by ●
- counterweights, so that it is articulated and can assume any position on the ground
 - a spring, so that it returns to the neutral position on the ground
 - a damper, so that it is held still while the aircraft is on the ground
- 295.** An aeroplane must hold its lift while the airspeed is reduced. What are the two ways of doing it?
- decrease the angle of attack or change the wing geometry by retracting slats and flaps
 - increase the angle of attack or retract the slats and flaps
 - increase the angle of attack or change the wing geometry by extending slats and flaps
- 296.** An angle of attack indication shows the pilot ●
- the maximum angle of attack permitted for the wing
 - the margin remaining up to alpha max
 - the actual angle of attack measured by the alpha vane
- 297.** A pilot pulls the stick back past the critical angle and the aircraft loses lift. What is the cause?
- Loss of airflow over the upper surface of the wing
 - Loss of airflow under the lower surface of the wing
 - An increase of air density over the wing
- 298.** On an aeroplane the approaching stall speed is not immediately noticeable through initial buffet. What must be installed? ●
- An angle of attack indicator
 - A stall warning system
 - An altitude alert system
- 299.** The purpose of the stall warning system ●
- Preventing the control column from being pulled back
 - Alerting the crew to the hazardous situation of a stall
 - Reducing the angle of attack automatically
- 300.** Where errors in the SWMCS are reported ●
- To EICAS and the GPWC
 - To EICAS and the CMC
 - To the CMC and the FCUs
- 301.** A technician wants to check the SWMCS with the CMC. When can this be done? ●
- In the air only
 - Both on the ground and in the air
 - On the ground only

- 302.** The speed that the SWMCS calculates as the stick shaker speed
- VMO, displayed as maximum operating speed
 - VBFLO, displayed as minimum operating speed
 - VBFLO, displayed as minimum manoeuvre speed
- 303.** The users of the pitch limit calculated by the SWMCS ●
- EICAS and the GPWC
 - The CMC and the IRS
 - EICAS and the FCUs
- 304.** The nature of the maximum angle of attack used by the SWMC ●
- It is a fixed value set at build
 - It is a variable value
 - It is selected by the crew before each flight
- 305.** Where the SWMC gets the flap position it needs to calculate the maximum angle of attack ●
- From the ADC
 - From the FCUs
 - From the IRS
- 306.** The unit that corrects the calculated pitch limit signal ●
- The ADC
 - The GPWC
 - The IRS
- 307.** The reason the stall warning management computer system is switched off below 40 kts ●
- Because the aircraft is then on the ground
 - Because the flap position is then not available from the FCUs
 - Because the stick shaker motors are then unpowered

LEVEL 3 — APPLY AND REASON

- 308.** During a walk round a technician finds the angle of attack vane hanging at a random angle. What should be concluded? ●
- The vane is seized and the sensor must be replaced
 - The vane counterweight has come adrift and the sensor must be rebalanced
 - This is normal, because the vane is force-free and can assume any position on the ground
- 309.** The quantity sensed by an angle of attack wind vane on the side of the fuselage, and the quantity sensed by a lift transducer below the forward wing surface ●
- The vane senses the angle of the air stream, the transducer a pressure difference between two rows of holes
 - The vane senses changes in the stagnation point, the transducer the angle of the air stream
 - The vane senses the angle of the air stream, the transducer changes in the stagnation point
- 310.** An aeroplane slows down in level flight with the slats and flaps left retracted. What happens to the angle of attack indication? ●
- It stays constant, because the lift in level flight does not change
 - It decreases, because less lift is being produced at the lower speed
 - It increases, and approaches alpha max as the speed falls

- 311.** A crew extends the slats and flaps while holding the same airspeed. Why does this help keep the angle of attack away from alpha max?
- The wing geometry is changed, so the required lift is produced at a smaller angle of attack
 - The alpha vane is biased by the slat position, so it reads a lower angle
 - The airspeed sensed by the wing is increased, so less lift has to be generated
- 312.** The angle of attack indication of an aeroplane is rising towards alpha max. What is about to happen at the wing? ●
- The airflow can no longer follow the upper surface of the wing and separation occurs
 - The airflow can no longer follow the lower surface of the wing and separation occurs
 - The airflow over the wing speeds up and the lift goes on rising
- 313.** In flight the maximum possible angle of attack for the flap position in use is exceeded. What happens? ●
- The flaps extend automatically
 - A motor shakes the control column
 - An amber lamp illuminates and a chime sounds
- 314.** An aeroplane is on stand with the APU supplying the busses and ground power also connected. What is the state of the wind vane heating?
- It starts as soon as the APU generator comes on line
 - It starts only when external power is selected
 - It does not start
- 315.** A technician holds the module switch at TEST and sees the warning lamp go off. What does this prove?
- Voltage and heating power are present
 - The shaker motor is unserviceable
 - The heating has been deactivated
- 316.** What the HTR OFF position does to the system
- It simulates FLIGHT status with the heating still energised
 - It simulates FLIGHT status with the heating deactivated
 - It isolates the system from the AC and the DC supply
- 317.** With the module in HTR OFF, how are the response values for flap position in the Operation Manual checked?
- By applying 20 V DC to the shaker test button on the back of the module
 - By turning the wind vane until the shaker motor vibrates
 - By moving the flap lever through each detent until the shaker motor vibrates
- 318.** The motor does not shake in TEST or HTR OFF mode. How can the shaker motor be checked directly?
- By pressing the TEST switch and applying 115 V AC
 - By pressing the shaker test button on the front panel and applying 28 V DC
 - By pressing the shaker test button on the back of the module and applying 20 V DC
- 319.** An aeroplane is taxiing at 30 kts with the pitch attitude unchanged. What is the state of the SWMCS? ●
- Active but with the shaker inhibited
 - Active and able to drive the shaker
 - Switched off

- 320.** The maximum angle of attack is exceeded in flight. What is the response of the digital system? ●
- Both stick shaker motors respond
 - The stick shaker is inhibited until the pitch limit is passed
 - Only one stick shaker motor responds
- 321.** The ADC fails and its M and CAS signals are lost. Which SWMC calculation loses its correction? ●
- The correction of the calculated pitch limit
 - The correction of the maximum angle of attack
 - The calculation of flap position
- 322.** An aeroplane is on its take-off roll at 50 kts with the pitch attitude unchanged. Is the stall warning management computer system active? ●
- No, because it becomes active only above 140 kts or above 5 deg of pitch angle
 - No, because it becomes active only above 140 kts and above 5 deg of pitch angle
 - Yes, because it becomes active above 40 kts whatever the pitch angle
- 323.** In the stall warning management computer, the value corrected by the signals from the ADC and the value corrected by the IRS ●
- M and CAS from the ADC correct the maximum angle of attack, the IRS corrects the pitch limit signal
 - M and CAS from the ADC correct the maximum angle of attack, the IRS corrects the flap position signal
 - M and CAS from the ADC correct the pitch limit signal, the IRS corrects the maximum angle of attack
- 324.** Which air data quantities does the selected ADC pass to the stall warning management computer?
- AOA, pitch angle, pitch rate and normal acceleration
 - AOA, TAS, CAS, Mach and VMO/MMO
 - Flap position, speedbrake lever position and Mach
- 325.** In flight the pitch angle becomes greater than the calculated pitch limit. What does this mean?
- There is the danger of a stall condition
 - The aircraft has exceeded the maximum operating speed
 - The minimum operating speed has been recalculated
- 326.** On the airspeed tape, where does the minimum operating speed lie in relation to the minimum manoeuvre speed?
- Below it
 - At the same point, as the two marks coincide
 - Above it

Answers — Angle of attack sensing and stall warning

- 283. c** Angle of attack is measured between the oncoming air stream and the wing chord. (L1, PDF p. 551)
- 284. c** Those three are the main factors given for lift. (L1, PDF p. 553)
- 285. a** Alpha max marks the onset of airflow separation. (L1, PDF p. 553)
- 286. a** The manual names the air stream to wing chord angle the angle of attack. (L1, PDF p. 555)
- 287. b** The manual calls the beginning of the loss of airflow the initial buffet, also named low speed buffet. (L1, PDF p. 555)
- 288. b** The extended digital system is called the stall warning management computer system. (L1, PDF p. 559)
- 289. b** Those three items are the printed make up of the SWMCS. (L1, PDF p. 559)
- 290. a** Lift rises with angle of attack, up to the individual maximum for the profile. (L2, PDF p. 551)
- 291. a** Beyond that value the flow separates from the surface. (L2, PDF p. 551)
- 292. a** The lift transducer works off the movement of the stagnation point; the other two describe the vane and the airstream tube. (L2, PDF p. 551)

- 293. c** The tube senses the angle as a pressure difference between its two rows of holes. (L2, PDF p. 551)
- 294. a** Counterweights balance the vane so nothing but the airflow moves it. (L2, PDF p. 551)
- 295. c** Either a larger angle of attack or more wing camber and area makes up for the lost airspeed. (L2, PDF p. 553)
- 296. c** The indication is of the actual measured angle. (L2, PDF p. 553)
- 297. a** Lift is lost because the airflow over the upper wing surface breaks down. (L2, PDF p. 555)
- 298. b** The manual makes a stall warning system mandatory exactly when the initial buffet does not give the warning. (L2, PDF p. 555)
- 299. b** The system alerts; it does not act on the controls. (L2, PDF p. 555)
- 300. b** Fault reporting goes to EICAS and to the central maintenance computer. (L2, PDF p. 559)
- 301. b** The CMC check is available in both phases. (L2, PDF p. 559)
- 302. b** VBFLO is the stick shaker speed and is shown as the minimum operating speed. (L2, PDF p. 559)
- 303. a** The pitch limit is calculated for EICAS and the ground proximity warning computer. (L2, PDF p. 559)
- 304. b** The manual states the maximum angle of attack varies rather than being fixed. (L2, PDF p. 559)
- 305. b** Flap position for the maximum angle of attack calculation comes from the flap control units. (L2, PDF p. 559)
- 306. c** Pitch limit correction is an IRS function. (L2, PDF p. 559)
- 307. a** Below that speed the aircraft is on the ground, which is why the system is switched off. (L2, PDF p. 559)
- 308. c** With no airflow and no restoring force, a serviceable vane rests wherever it happens to be. (L3, PDF p. 551)
- 309. c** The wind vane senses the angle of the air stream directly, while the lift transducer senses changes in the direction and intensity of the stagnation point. (L3, PDF p. 551)
- 310. c** With the wing geometry unchanged, holding the lift at a lower airspeed demands a larger angle of attack, which moves the indication toward alpha max. (L3, PDF p. 553)
- 311. a** Changing the wing geometry is the alternative to increasing the angle of attack for the same lift. (L3, PDF p. 553)
- 312. a** The indication shows the actual angle of attack, and at alpha max the airflow cannot follow the upper surface of the wing and separates. (L3, PDF p. 553)
- 313. b** Exceeding the flap dependent maximum angle drives the stick shaker motor. (L3, PDF p. 555)
- 314. c** Heating needs engine generator power, so with APU or external power it stays off. (L3, PDF p. 557)
- 315. a** The lamp going off in TEST is the indication that voltage and heating power are there. (L3, PDF p. 557)
- 316. b** HTR OFF simulates FLIGHT status for normal operation but removes the heating. (L3, PDF p. 557)
- 317. b** The wind vane is turned and the shaker motor must vibrate at the published response values. (L3, PDF p. 557)
- 318. c** The direct shaker check uses the button on the back of the module with 20 V DC applied. (L3, PDF p. 557)
- 319. c** At less than 40 kts the system is off, which is the normal ground case. (L3, PDF p. 559)
- 320. a** The digital system shakes both columns when the maximum angle of attack is exceeded. (L3, PDF p. 559)
- 321. b** M and CAS from the ADC correct the maximum angle of attack; the pitch limit is corrected by the IRS instead. (L3, PDF p. 559)
- 322. a** At 50 kts the system is above the switch off speed but neither activation condition is met, so it is not yet active. (L3, PDF p. 559)
- 323. a** The ADC signals M and CAS correct the maximum angle of attack, while the pitch limit signal is corrected by the IRS. (L3, PDF p. 559)
- 324. b** The selected air data computer supplies those air data values; the pitch and acceleration terms come from the IRS. (L3, PDF p. 560)
- 325. a** Passing the calculated pitch limit is the printed warning of a stall danger. (L3, PDF p. 561)
- 326. a** The tape shows the minimum operating speed lower down than the minimum manoeuvre speed. (L3, PDF p. 562)